

Untitled hardware project

Class drone	Feasibility conditional	BoM Readiness Ready	Score 7.7
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Executive Summary

Metric	Value	Unit	Confidence	Notes
Flight endurance	40	minutes	medium	Target endurance for prosumer cinematography flight profile.
Estimated CAPEX	0	GBP	low	No CAPEX claim admitted yet; supplier-backed BoM evidence is required before costs appear.
Estimated annual OPEX	0	GBP/year	low	No OPEX claim admitted yet because CAPEX has no sourced BoM basis.

Brief Intake Gate

brief_intake_review_required

5/6 intake areas pass; 1 review, 0 blocked. Class drone (high); requirements 1; numeric facts 1; requirement trace 1/1 covered; scratch grammar yes.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
brief_text	pass	1	138 non-whitespace character(s) in the source brief.	none	Brief text is present.
requirement_quantification	review	0.5	1 quantified requirement(s), 1 numeric fact(s).	none	Proceed only as assumption-heavy review; ask for more quantified requirements before treating outputs as stable.
product_class_selection	pass	1	Selected drone with high confidence.	none	Product class selection is strong enough for scratch architecture.
scratch_design_support	pass	1	drone has a deep scratch architecture grammar.	none	Use scratch universal architecture flow for this brief.
requirement_trace_seed	pass	1	1 covered, 0 partial, 0 uncovered requirement trace row(s).	none	Extracted requirements have trace seeds into the architecture.
assumption_boundary	pass	1	1 brief assumption note(s); no-quantified-default path: no.	none	Assumption boundary is explicit enough for first-pass architecture.

Brief Clarification Plan

clarification_recommended

4 clarification question(s): 0 required, 3 recommended, 1 optional. Architecture blockers 0. Next question: target_metrics:1.

#	Kind	Priority	Status	Question	Expected Answer	Example	Blocks Architecture
1	target_metrics	major	recommended	What are the top two or three quantified performance targets?	Metric name, value and unit for each target.	40 minute flight endurance, 4K payload, 2 kg maximum take-off mass.	no
2	operating_envelope	major	recommended	What operating envelope should the design satisfy?	A short list of dimensions, duty cycle, environment, runtime, capacity or output constraints.	Outdoor wind tolerance, 40 minute mission, removable battery, transportable case.	no
3	interfaces_and_integration	major	recommended	What external systems, users or infrastructure must this design interface with?	List each external interface and any protocol, voltage, fluid, data or mechanical constraint.	Remote controller, GPS, camera gimbal, battery charger and payload mount.	no
4	compliance_and_risk	minor	optional	Are there named standards, hazards or certification targets that must be treated as in-scope?	Named standard, jurisdiction, hazard or certification target.	Open category mass limits, radio approvals, battery transport requirements.	no

Architecture Admission Gate

architecture_generation_review_required

4/6 admission areas pass; 2 review, 0 blocked. Can use for review: yes; can proceed to BoM: no. Source scratch_universal_architecture; next action: Treat generated architecture as review-only until recommended clarification questions are answered..

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
brief_intake_clearance	review	0.83	5/6 intake areas pass; class drone (high); 1 quantified requirement(s).	none	Treat generated architecture as review-only until recommended clarification questions are answered.
clarification_clearance	review	1	4 clarification question(s): 0 required, 3 recommended, 0 architecture-blocking.	none	Keep the architecture labelled as review-only until recommended brief clarifications are answered.
class_grammar_clearance	pass	1	Product class drone; scratch universal grammar supported: yes.	none	Route this class through the scratch universal module grammar.
scratch_stage_clearance	pass	1	Architecture stage passed; source scratch_universal_architecture; modules 9; submodules 26; components 110.	none	Architecture was produced by the scratch universal architecture stage.
architecture_readiness_boundary	pass	1	Architecture ready for BoM: yes; blocking issues 0; required links 3/3.	none	Deterministic architecture readiness checks passed; sourcing work may begin only with explicit evidence.
bom_provenance_hold	pass	1	0 admitted sourcing evidence row(s), 0 source-backed BoM claim(s), 0 provenance violation(s).	none	BoM fields remain behind the explicit sourcing-evidence boundary.

Stage Integrity Gate

stage_trace_accepted

5/5 stage-integrity areas pass; 0 review, 0 blocked. Stages 8/8; ordered: yes; architecture source scratch_universal_architecture; admitted priced lines 0; provenance violations 0.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
stage_sequence	pass	1	8/8 expected compiler stages are present in order.	none	Stage sequence matches the scratch compiler contract.
stage_payloads	pass	1	8/8 stages have metrics; 8/8 have evidence or explicit limitation; 8/8 have limitations.	none	Every stage carries metrics, evidence or explicit limitation, and limitations.
scratch_architecture_source	pass	1	Product class drone; architecture source scratch_universal_architecture; scratch grammar supported: yes.	none	Architecture came from the scratch universal grammar.
architecture_before_sourcing	pass	1	Architecture ready: yes; readiness stage passed; sourcing stage warning.	none	Sourcing stage is downstream of architecture readiness and has a consistent status.
sourcing_provenance_boundary	pass	1	0 admitted source evidence row(s), 0 admitted priced line(s), 0 provenance violation(s).	none	Supplier, manufacturer, MPN and cost fields stay behind admitted sourcing evidence.

Scratch Lineage Gate

scratch_lineage_clean

6/6 lineage areas pass; 0 review, 0 blocked. Architecture source scratch_universal_architecture; design refs 237; source refs 0; forbidden refs 0; forbidden stage mentions 0; chain-v2 benchmark no (prototype internal depth target).

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture_source	pass	1	Product class drone; architecture source scratch_universal_architecture; scratch grammar supported: yes.	none	Design content came from the scratch universal architecture stage.
design_provenance_refs	pass	1	463 provenance ref(s); 237 design/architecture ref(s); 0 invalid kind(s); 0 forbidden ref(s).	none	All design provenance references are in the allowed scratch provenance vocabulary.
bom_candidate_provenance	pass	1	110 candidate BoM line(s); 0 admitted source evidence row(s); 0 unprovenanced supplier/manufacturer/MPN/cost claim(s).	none	Candidate BoM fields remain unpriced or explicitly source-backed.
source_evidence_boundary	pass	1	0 source provenance ref(s); 0 outside admitted evidence boundary; 0 inside architecture content.	none	Source refs appear only in admitted source evidence, source ledger entries or matching source-backed BoM lines.
chain_v2_quarantine	pass	1	0 forbidden provenance ref(s); 0 forbidden stage-trace mention(s).	none	No chain-v2/state.json/modifier lineage appears inside generated design content or stage traces.
benchmark_isolation	pass	1	Benchmark source: prototype internal depth target; chain benchmark used: no; policy: Benchmark uses aggregate counts only. Chain-v2 component names, prose, design content, prices and part numbers are not imported into the scratch generator.	none	Benchmark does not use chain-v2 input.

Architecture Freeze Gate

architecture_freeze_review_required

1/6 freeze areas pass; 5 review, 0 blocked. Modules/submodules/components 9/26/110. Structurally ready for sourcing yes; independent review accepted no. Module engineering module_engineering_review_required; interfaces interface_review_ready; requirements coverage_review_ready; next action: Keep architecture labelled review-ready until admission, stage trace and scratch lineage are all clean..

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
admission_and_lineage	review	0.88	Architecture admission architecture_generation_review_required; stage integrity stage_trace_accepted; scratch lineage scratch_lineage_clean.	none	Keep architecture labelled review-ready until admission, stage trace and scratch lineage are all clean.
module_structure	review	0.74	Component allocation allocation_ready; module engineering module_engineering_review_required; 0/9 modules pass; 5 modules still have critical sourcing blocks.	none	Treat the architecture as review-ready until module review questions and critical sourcing blocks close.
interface_contracts	review	1	Interface verification interface_review_ready; 3/3 required interfaces have carriers; 0 accepted.	none	Collect or accept interface-review evidence before calling the architecture fully frozen.
requirement_coverage	review	1	Requirement coverage coverage_review_ready; 1/1 requirements structurally ready or accepted; accepted ratio 0.	none	Requirement coverage is structurally ready; collect accepted reviewer evidence before full freeze.
engineering_review_state	review	0.67	Pre-BoM engineering engineering_review_ready; 4/6 areas pass; calculations needing review 1; assumptions needing review 16.	none	Keep architecture labelled review-ready until calculations, assumptions and verification evidence are accepted.
sourcing_boundary	pass	1	110 candidate BoM lines; 0 admitted sourcing evidence rows; 0 provenance violation(s); 32 critical missing source claim(s).	none	Sourcing and BoM claims remain behind explicit admitted source evidence.

Architecture Freeze Closure Plan

architecture_freeze_review_required

5 closure row(s): 5 ready, 0 blocked. Queues S/E/V/R 1/1/2/1; next row freeze-close:1:admission_and_lineage. Structurally ready for sourcing yes; independent review accepted no.

#	Area	Queue	Priority	Status	Required Evidence	Acceptance Criteria	Input Artifacts
1	admission_and_lineage	architecture_review	minor	ready	Keep architecture labelled review-ready until admission, stage trace and scratch lineage are all clean.	Architecture admission verdict is architecture_generation_admitted. Stage integrity verdict is stage_trace_accepted. Scratch lineage verdict is scratch_lineage_clean.	*.architecture-admission-gate.csv; *.stage-integrity-gate.csv; *.scratch-lineage-gate.csv
2	module_structure	sourcing_intake	major	ready	Treat the architecture as review-ready until module review questions and critical sourcing blocks close.	Module engineering verdict is module_engineering_ready. No module has blocked allocation or unresolved critical sourcing blockers. Every module review question is accepted, revised or explicitly deferred.	*.module-engineering-gate.csv; *.component-allocation-gate.csv; *.engineering-review-pack.csv; *.sourcing-intake-template.csv
3	interface_contracts	verification_intake	minor	ready	Collect or accept interface-review evidence before calling the architecture fully frozen.	Interface verification verdict is accepted_interfaces. Every required interface has endpoint carriers and accepted reviewer evidence.	*.interface-contracts.json; *.interface-verification-gate.csv; *.verification-intake-template.csv
4	requirement_coverage	verification_intake	minor	ready	Requirement coverage is structurally ready; collect accepted reviewer evidence before full freeze.	Requirement coverage verdict is accepted_evidence. Every parsed requirement has architecture links and accepted verification evidence.	*.requirement-coverage-gate.csv; *.engineering-assurance-matrix.csv; *.verification-intake-template.csv

#	Area	Queue	Priority	Status	Required Evidence	Acceptance Criteria	Input Artifacts
5	engineering_review_state	engineering_review	major	ready	Keep architecture labelled review-ready until calculations, assumptions and verification evidence are accepted.	Pre-BoM engineering verdict is engineering_accepted. Calculations, assumptions and verification activities are accepted or explicitly deferred.	*.pre-bom-engineering-gate.csv; *.engineering-calculations.csv; *.engineering-assumptions.csv; *.verification-plan.csv

Component Candidate Gate

component_candidates_review_required

6/7 candidate areas pass; 1 review, 0 blocked. Candidate lines 110; critical 8/8; worklist rows 110; duplicate groups 0; provenance violations 0; ready for sourcing no.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
candidate_presence	pass	1	110 candidate BoM line(s), 110 distinct component identity key(s).	none	Component candidates exist for downstream sourcing review.
candidate_identity	review	0.99	109/110 candidate line(s) have non-generic identity, description and role context.	actuation_kinematics_esc_stack_esc: weak candidate identity for esc.	Rename weak component candidates and add role context before sourcing them.
quantity_basis	pass	1	110/110 candidate line(s) have positive quantity and unit.	none	Candidate quantities have positive values and explicit units.
critical_candidate_coverage	pass	1	8/8 class-critical part(s) are represented by candidate lines; 8 critical BoM line(s).	none	Class-critical parts have candidate lines before sourcing begins.
duplicate_identity_review	pass	1	0 duplicate component identity group(s), 0 involving critical lines.	none	No duplicate component identities need sourcing review.
sourcing_worklist_alignment	pass	1	110/110 unpriced candidate line(s) appear in the sourcing worklist.	none	Every unpriced candidate line is visible in the sourcing worklist.
provenance_boundary	pass	1	0 source evidence row(s), 0 source-backed claim(s), 0 provenance violation(s).	none	Candidate sourcing fields remain clean unless admitted evidence supplies them.

Sourcing Authorization Gate

sourcing_authorization_review_required

4/6 sourcing-authorization areas pass; 2 review, 0 blocked. Critical intake 8/8; full intake 110/110; admitted evidence 0; rejected evidence 0; provenance violations 0; authorized no.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture_authorized	review	0.5	architecture_generation_review_required; review yes, BoM no.	none	Keep sourcing as review-only until architecture admission is fully cleared.
component_candidate_authorization	review	0.86	component_candidates_review_required; 110 candidate line(s), 110 sourcing-worklist row(s), 0 provenance violation(s).	none	Resolve candidate review rows before admitting sourcing evidence.
critical_intake_scope	pass	1	8/8 critical unpriced line(s) appear in the critical sourcing intake template.	none	Critical sourcing intake scope is complete.
full_worklist_scope	pass	1	110/110 total unpriced line(s) appear in the full sourcing intake template.	none	Full sourcing intake scope matches the unpriced worklist.
evidence_admission_boundary	pass	1	0 admitted source evidence row(s), 0 rejected source evidence row(s).	none	No rejected sourcing evidence is waiting for repair.
provenance_boundary	pass	1	0 source-backed BoM claim(s), 0 provenance violation(s).	none	Supplier, manufacturer, MPN, lead-time and cost fields remain behind admitted evidence.

BoM Admission Gate

candidate_bom_review_required

Display mode candidate_only. 1/7 BoM-admission areas pass; 6 review, 0 blocked. Critical priced 0/8; source-backed claims 0; provenance violations 0. Candidate BoM yes; priced BoM no; procurement use no.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
sourcing_authorization	review	0.67	sourcing_authorization_review_required ; authorized no; 4/6 areas pass.	none	Keep BoM admission review-only until sourcing authorization clears.
candidate_bom_scope	pass	1	110 candidate BoM line(s), 8 critical line(s).	none	Candidate BoM scope exists for sourcing and evidence collection.
source_admission_state	review	0	0 admitted source row(s), 0 rejected source row(s), 0/110 priced line(s).	none	No source rows are admitted; keep this as a candidate-only BoM.
critical_bom_admission	review	0	0/8 critical line(s) have admitted unit costs; 8 critical line(s) remain unpriced.	none	Start with critical sourcing intake before treating any BoM cost as admitted.
source_reference_quality	review	0	no_sourcing_evidence; 0/0 source-quality rows pass; 0 protocol fixture row(s).	none	No source evidence is present; keep the BoM candidate-only.
provenance_boundary	review	0.94	0 source-backed claim(s), 32 critical missing-source claim(s), 0 provenance violation(s).	none	Missing critical source claims are acceptable only in candidate-only mode; collect evidence before priced BoM use.
display_mode	review	0.25	Display mode will be candidate_only.	none	Show candidate BoM as a sourcing queue with supplier, MPN and cost fields empty.

Report Readiness Gate

architecture_review_ready

5/6 sections at or above 8. Mean score 7.67. Architecture ready: yes. Critical unpriced lines: 8. Verification accepted: 0/19; unaccepted: 19. Verification acceptance ratio: 0.

- bom score 1 is below 8.
- 8 critical BoM line(s) lack source-backed cost evidence.
- 5 verification activity/activities remain blocked.
- 0/19 non-sourcing verification activity/activities have accepted reviewer evidence.

Section	Score	Target	B/M/m Issues	Action
executive_summary	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.
brief_requirements	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.
design_modules	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.
bom	1	below target	8/1/0	Admit source-backed evidence for every critical BoM line through sourcing intake. Increase priced-line coverage with admissible supplier evidence, not estimates.
feasibility_notes	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.
sources_references	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.

Pre-BoM Engineering Gate

engineering_review_ready

4/6 areas pass; 2 review, 0 blocked. Modules/submodules/components: 9/26/110. Calculations within envelope 1/2; calculation blockers 0. Assumption review rows 16; architecture blockers 0.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture_readiness	pass	1	9 modules, 26 submodules and 110 component words pass architecture readiness.	none	Architecture is structurally ready for engineering review.
component_allocation	pass	1	26/26 submodules ready; 8/8 critical parts allocated; 0 duplicate component groups.	none	Component candidates are allocated well enough for engineering review.
interface_verification	pass	1	3/3 required interfaces have endpoint carriers; 0 accepted with evidence.	none	Required interfaces are structurally ready for reviewer evidence.
requirement_coverage	pass	1	1/1 requirements are structurally ready or accepted; accepted evidence ratio 0.	none	Requirements are structurally ready for reviewer evidence.
calculation_envelope	review	0.5	1/2 calculations are within envelope; 1 need review; 0 block.	none	Attach reviewer evidence to calculations marked needs_review.
engineering_assumptions	review	0.2	4/20 assumptions are brief-supported or model-present; 16 need review/source evidence; 0 block architecture.	none	Review assumptions and keep BoM/source-required assumptions out of trusted claims until evidence is admitted.

BoM Costing Gate

costing_not_started

1/7 costing areas pass; 0 review, 6 blocked. Critical priced lines 0/8; unpriced critical lines 8. Source-backed claims 0; provenance violations 0; sourcing evidence 0/0 production-ready, 0 protocol. Source quality: no_sourcing_evidence (0/0 pass, 0 placeholder URL, 0 quote-anchored). BoM total £0; CAPEX £0.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
sourcing_admission	blocked	0	0/110 BoM lines have admitted pricing; rejected source records 0.	structure_containment_carbon_airframe_carbon_airframe: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. energy_storage_source_flight_battery_pack_flight_battery_pack: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. actuation_kinematics_motor_sets_brushless_motors: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. actuation_kinematics_esc_stack_esc: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. actuation_kinematics_propeller_sets_propeller_set: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. plus 3 more	Admit source-backed supplier, manufacturer, MPN and unit-cost evidence before costing review.
critical_line_pricing	blocked	0	0/8 critical lines have admitted unit costs.	structure_containment_carbon_airframe_carbon_airframe: Carbon airframe energy_storage_source_flight_battery_pack_flight_battery_pack: Flight battery pack actuation_kinematics_motor_sets_brushless_motors: Brushless motors actuation_kinematics_esc_stack_esc: ESC actuation_kinematics_propeller_sets_propeller_set: Propeller set plus 3 more	Price every critical line from admitted source evidence.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
source_backed_provenance	blocked	0.94	0 source-backed claims; 32 critical missing-source claims; 0 provenance violations.	structure_containment_carbon_airframe_carbon_airframe/supplier: Admit source-backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass. structure_containment_carbon_airframe_carbon_airframe/manufacturer: Admit source-backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass. structure_containment_carbon_airframe_carbon_airframe/mpn: Admit source-backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass. structure_containment_carbon_airframe_carbon_airframe/unit_cost_gbp: Admit source-backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass. energy_storage_source_flight_battery_pack_flight_battery_pack/supplier: Admit source-backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass. plus 27 more	Keep supplier, manufacturer, MPN, unit cost and lead-time fields empty until source-backed evidence is admitted.
component_identity	pass	1	0 duplicate component identity group(s); 0 include critical lines.	none	Component identities are unique enough for line-level costing.
sourcing_evidence_authenticity	blocked	0	0/0 sourcing evidence rows are production-ready; 0 protocol fixture, 0 review/missing metadata.	none	Add sourcing evidence rows before BoM costing can start.
source_reference_quality	blocked	0	0/0 sourcing refs pass quality checks; 0 placeholder URL,	none	Add source references before costing review.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
			0 quote-anchored, 0 fresh timestamp.		
cost_totals	blocked	0	BoM total 0 GBP; CAPEX 0 GBP; annual OPEX 0 GBP.	8 critical line(s) are unpriced, so totals are incomplete.	Do not use cost totals until every critical line has an admitted unit cost.

Evidence Acquisition Plan

32 missing-evidence row(s): 11 sourcing, 21 verification. Ready 29; blocked 0. Blocks BoM 11; blocks publish 32. Next row: acquire:4:assumption:metric:capex_gbp.

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
1	sourcing	blocker	open	readiness:2	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
2	verification	blocker	open	readiness:3	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
3	verification	blocker	open	readiness:4	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
4	sourcing	blocker	ready_for_intake	assumption:metric:capex_gbp	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated;	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						sourceGrade; evidence.ref; evidence.quote; retrievedAt		
5	sourcing	blocker	ready_for_intake	assumption:metric:opex_annual_gbp	supplier_catalogue_or_quote	componentWorldId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
6	sourcing	blocker	ready_for_intake	carbon_airframe	supplier_catalogue_or_quote	componentWorldId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
7	sourcing	blocker	ready_for_intake	flight_battery_pack	supplier_catalogue_or_quote	componentWorldId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade;	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						evidence.ref ; evidence.quote; retrievedAt		
8	sourcing	blocker	ready_for_in take	brushless_ motors	supplier_cat alogue_or_q uote	component WordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref ; evidence.quote; retrievedAt	LLM- estimated cost; benchmark average without supplier evidence; class-pack default; test- fixture:// source	yes
9	sourcing	blocker	ready_for_in take	esc	supplier_cat alogue_or_q uote	component WordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref ; evidence.quote; retrievedAt	LLM- estimated cost; benchmark average without supplier evidence; class-pack default; test- fixture:// source	yes
10	sourcing	blocker	ready_for_in take	propeller_se t	supplier_cat alogue_or_q uote	component WordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref ;	LLM- estimated cost; benchmark average without supplier evidence; class-pack default; test- fixture:// source	yes

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						evidence.quote; retrievedAt		
11	sourcing	blocker	ready_for_intake	flight_controller	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
12	sourcing	blocker	ready_for_intake	radio_and_gnss_link	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
13	sourcing	blocker	ready_for_intake	4k_camera_payload	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						ote; retrievedAt		
14	verification	major	ready_for_in take	design_review:structure_containment	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; ; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
15	verification	major	ready_for_in take	design_review:energy_storage_source	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; ; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
16	verification	major	ready_for_in take	calculation:energy_storage_source	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; ; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
17	verification	major	ready_for_in take	design_review:actuation_kinematics	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; ; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
18	verification	major	ready_for_intake	calculation: actuation_kinematics	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
19	verification	major	ready_for_intake	design_review:control_compute_communication	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
20	verification	major	ready_for_intake	calculation: control_compute_communication	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
21	verification	major	ready_for_intake	design_review:sensing_instrumentation	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
22	verification	major	ready_for_intake	design_review:power_distribution	named_engineering_review	activityId; evidenceKind; reviewerNa	anonymous reviewer; unchecked model	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						me; verdict; evidenceRef; evidenceNote; reviewedAt	assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	
23	verification	major	ready_for_intake	design_review:safety_protection	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
24	verification	major	ready_for_intake	design_review:maintenance_serviceability	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
25	verification	major	ready_for_intake	design_review:hmi ergonomics	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
26	verification	major	ready_for_intake	interface_review:energy_storage_source:actuation_kinematics:power_bus	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNo	anonymous reviewer; unchecked model assertion; test- fixture:// review; source-	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						te; reviewedAt	evidence claims that belong in sourcing intake	
27	verification	major	ready_for_in take	interface_re view:control _compute_c ommunicati on:actuation _kinematics :motor_driv e	named_engi neering_revi ew	activityId; evidenceKin d; reviewerNa me; verdict; evidenceRef ; evidenceNo te; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
28	verification	major	ready_for_in take	interface_re view:control _compute_c ommunicati on:sensing_ instrumenta tion:sensor_ bus	named_engi neering_revi ew	activityId; evidenceKin d; reviewerNa me; verdict; evidenceRef ; evidenceNo te; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
29	verification	major	ready_for_in take	compliance _review:UK- UAS	named_engi neering_revi ew	activityId; evidenceKin d; reviewerNa me; verdict; evidenceRef ; evidenceNo te; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
30	verification	major	ready_for_intake	compliance_review:CE-EMC	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
31	verification	major	ready_for_intake	risk_review:1	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
32	verification	major	ready_for_intake	risk_review:2	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no

Document Trust Gate

evidence_blocked

1/7 trust areas pass; 3 review, 3 blocked. Architecture ready: yes. Claim gate: evidence_blocked. Reviewer evidence 0/19. Source-backed BoM claims 0; critical missing BoM claims 32; provenance violations 0. Evidence authenticity: no_evidence (0/0 production-ready, 0 protocol).

- bom: score 1 below 8
- architecture design has 0 blocked/source-required claim(s) and 35 review-required claim(s).
- engineering math has 2 blocked/source-required claim(s) and 3 review-required claim(s).
- bom sourcing has 40 blocked/source-required claim(s) and 102 review-required claim(s).
- compliance risk has 0 blocked/source-required claim(s) and 4 review-required claim(s).
- structure_containment_carbon_airframe_carbon_airframe/supplier: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
- structure_containment_carbon_airframe_carbon_airframe/manufacturer: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
- structure_containment_carbon_airframe_carbon_airframe/mpn: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.

- structure_containment_carbon_airframe_carbon_airframe/unit_cost_gbp: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
- energy_storage_source_flight_battery_pack_flight_battery_pack/supplier: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
- energy_storage_source_flight_battery_pack_flight_battery_pack/manufacturer: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
- energy_storage_source_flight_battery_pack_flight_battery_pack/mpn: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture_readiness	pass	1	9 modules, 26 submodules and 3 required links pass deterministic architecture gates.	none	No deterministic architecture blocker remains; continue with reviewer evidence before treating it as accepted.
section_scores	blocked	0.83	5/6 scored sections are at or above 8; mean 7.67.	bom: score 1 below 8	Repair below-target sections through the specific score-ledger actions.
claim_evidence	blocked	0.01	1/187 claims pass; 42 blocked/source-required, 144 review-required.	architecture design has 0 blocked/source-required claim(s) and 35 review-required claim(s). engineering math has 2 blocked/source-required claim(s) and 3 review-required claim(s). bom sourcing has 40 blocked/source-required claim(s) and 102 review-required claim(s). compliance risk has 0 blocked/source-required claim(s) and 4 review-required claim(s).	Accept, correct or reject generated module and submodule allocation claims through engineering review. Accept, correct or reject calculated metrics and engineering envelopes through reviewer evidence. Admit source-backed supplier, manufacturer, MPN, unit-cost and lead-time evidence for required BoM claims. Accept, correct or reject class-pack regulatory and risk claims through compliance/domain review.
requirement_assurance	review	0	0/1 requirements have accepted assurance; 1 need review, 0 blocked, 0 unlinked.	none	Collect reviewer evidence for requirement assurance rows before publication.
reviewer_evidence	review	0	0/19 non-sourcing verification activities accepted; 0 rejected, 0 deferred, 19 pending/blocked.	none	Collect accepted reviewer evidence for every non-sourcing verification activity.
bom_provenance	blocked	0.94	0 source-backed BoM claims, 32 critical missing-source claims, 0 provenance violations.	structure_containment_carbon_airframe_carbon_airframe/supplier: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. structure_containment_carbon_airframe_carbon_airframe/manufacturer:	Admit source-backed evidence for critical BoM supplier, manufacturer, MPN and cost claims.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
				Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. structure_containment_carbon_airframe_carbon_airframe/mpn: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. structure_containment_carbon_airframe_carbon_airframe/unit_cost_gbp: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. energy_storage_source_flight_battery_pack_flight_battery_pack/supplier: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. plus 27 more	

evidence_authenticity	review	0	0/0 evidence row(s) are production-ready; 0 protocol fixture, 0 review/missing metadata.	none	Admit source and reviewer evidence before trusting the document.
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Evidence Authenticity Gate

no_evidence

0/0 evidence rows are production-ready. Protocol fixtures 0; review-required 0; missing metadata 0. External URLs 0; internal references 0; local files 0; unknown references 0.

Kind	Subject	Reference Class	Status	Ref	Reason	Required Action
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Evidence Replacement Plan

No evidence replacement rows remain for the current authenticity verdict: no_evidence.

Trust Repair Plan

6 package(s): 6 ready, 0 waiting. Next package: sourcing_intake. Closure rows 40; trust blockers 45. Trust verdict: evidence_blocked.

#	Package	Status	Gate Areas	Closure Rows	Trust Blockers	Required Inputs	Source Artifacts	Exit Criteria
2	Admit Source-Backed BoM Claims	ready	claim_evidence; bom_provenance	11	36	Supplier, manufacturer, MPN, unit cost, lead time, retrieved-at timestamp and source reference for critical BoM lines. Rejected or duplicate component identities resolved before pricing shared items.	*.sourcing-intake-template.csv; *.sourcing-pack.csv; *.bom-provenance-manifest.csv; *.component-identity.csv	BoM provenance has zero critical missing-source claims. Claim Evidence Gate BoM sourcing area is no longer evidence_blocked.
3	Resolve Engineering Review Questions	ready	claim_evidence; requirement_assurance	7	4	Accepted/corrected/rejected answers to review-pack questions. Calculation and assumption decisions tied to reviewer evidence references.	*.engineering-review-pack.csv; *.engineering-assurance-matrix.csv; *.engineering-assumptions.csv; *.engineering-calculations.csv	Requirement assurance rows have accepted or explicitly deferred engineering review evidence. Generated architecture and calculation claims are accepted, corrected or rejected.
4	Admit Reviewer Verification Evidence	ready	reviewer_evidence	21	0	Accepted, rejected or deferred verification evidence for every non-source verification activity.	*.verification-intake-template.csv; *.verification-plan.csv; *.verification-ledger.csv	Every non-source verification activity has accepted reviewer evidence or an explicit defer decision.
5	Validate Evidence Authenticity	ready	evidence_authenticity	0	0	Production supplier/catalogue URLs for BoM source records. Named reviewer evidence references that are not protocol fixtures.	*.evidence-authenticity-gate.csv; *.sourcing-ledger.csv; *.verification-ledger.csv; *.claim-ledger.csv	Evidence Authenticity Gate verdict is production_ready. No protocol_fixture, unknown_reference or missing_metadata evidence

#	Package	Status	Gate Areas	Closure Rows	Trust Blockers	Required Inputs	Source Artifacts	Exit Criteria
								rows remain.
6	Review Compliance And Risk Claims	ready	claim_evidence	0	4	Compliance -domain review of class-pack standards, hazards and mitigation claims.	*.claim-ledger.csv; *.engineering-review-pack.csv	Compliance and risk claims are accepted, corrected or rejected by a domain reviewer.
7	Repair Residual Section Scores	ready	section_scores	1	1	A rerun after evidence intake, plus score-ledger fixes for any section still below target.	*.score-ledger.csv; *.readiness-gate.csv; *.document-trust-gate.csv	All scored sections are at or above target and Document Trust Gate is no longer blocked by scoring.

Claim Ledger

187 claims: 1 brief-supplied, 0 reviewer-accepted, 0 source-backed, 3 calculated needing review, 141 generated needing review, 42 source-required, 0 blocked. Blocking: 42 BoM, 185 publishable.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
bom_source_field	source_required	assumption	ESC / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	ESC / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	ESC / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	ESC / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Brushless motors / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Brushless motors / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
								before BoM can pass.
bom_source_field	source_required	assumption	Brushless motors / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Brushless motors / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Propeller set / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Propeller set / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Propeller set / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Propeller set / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
								cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Flight controller / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Flight controller / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Flight controller / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Flight controller / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Radio and GNSS link / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Radio and GNSS link / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/ma

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
								nufacturer/ MPN/unit- cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Radio and GNSS link / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/ MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Radio and GNSS link / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/ MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Flight battery pack / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/ MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Flight battery pack / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/ MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Flight battery pack / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/ MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Flight battery	unit_cost_gbp is	none	yes	yes	Admit source-

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
			pack / unit_cost_gbp	required but not source-backed.				backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	4K camera payload / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	4K camera payload / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	4K camera payload / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	4K camera payload / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Carbon airframe / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
bom_source_field	source_required	assumption	Carbon airframe / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Carbon airframe / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Carbon airframe / unit_cost_g bp	unit_cost_g bp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
component_candidate	source_required	generated_model	ESC	ESC is an architecture-derived component candidate with quantity 4 each.	model:scratch_universal_architecture.drone	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	Brushless motors	Brushless motors is an architecture-derived component candidate with quantity 4 each.	model:scratch_universal_architecture.drone	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	Propeller set	Propeller set is an architecture	model:scratch_universal_architecture.drone	yes	yes	Admit source-backed

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
component_candidate	source_required	generated_model	Flight controller	-derived component candidate with quantity 1 candidate.	l_architecture.drone			supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
				Flight controller is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.drone	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
				Radio and GNSS link	model:scratch_universal_architecture.drone	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
				Flight battery pack	model:scratch_universal_architecture.drone	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
				4K camera payload	model:scratch_universal_architecture.drone	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
								sourcing intake.
component_candidate	source_required	generated_model	Carbon airframe	Carbon airframe is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.drone	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
headline_metric	source_required	assumption	Estimated CAPEX	Estimated CAPEX = 0 GBP.	formula:capex=sum(bom.totalCostGbp); class_pack:drone.class_pack	yes	yes	Admit source-backed BoM costs before treating this cost metric as a claim.
headline_metric	source_required	assumption	Estimated annual OPEX	Estimated annual OPEX = 0 GBP/year.	formula:opex=capex*0.04	yes	yes	Admit source-backed BoM costs before treating this cost metric as a claim.
component_candidate	generated_needs_review	generated_model	current telemetry line	current telemetry line is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.drone	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	ESC conformal coating	ESC conformal coating is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.drone	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	ESC cooling plate	ESC cooling plate is an architecture-derived component candidate with	model:scratch_universal_architecture.drone	no	yes	Review whether this is a unique physical item, shared component or separate

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
				quantity 1 candidate.				installation instance.
component_candidate	generated_needs_review	generated_model	PWM signal harness	PWM signal harness is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.drone	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	motor bearing shield	motor bearing shield is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.drone	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	motor mounting screw set	motor mounting screw set is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.drone	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	motor phase lead set	motor phase lead set is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.drone	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	propeller hub adapter	propeller hub adapter is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.drone	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	propeller lock nut	propeller lock nut is an architecture-derived	model:scratch_universal_architecture.drone	no	yes	Review whether this is a unique physical item, shared

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
				component candidate with quantity 1 candidate.				component or separate installation instance.
component_candidate	generated_needs_review	generated_model	spare propeller pouch	spare propeller pouch is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.drone	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	barometer foam cover	barometer foam cover is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.drone	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	failsafe firmware profile	failsafe firmware profile is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.drone	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	IMU vibration isolation mount	IMU vibration isolation mount is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.drone	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	SD card flight logger	SD card flight logger is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.drone	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
component_candidate	generated_needs_review	generated_model	antenna diversity mount	antenna diversity mount is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.drone	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	GNSS antenna mast	GNSS antenna mast is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.drone	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	RC receiver	RC receiver is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.drone	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	telemetry radio module	telemetry radio module is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.drone	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	HDMI micro cable	HDMI micro cable is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.drone	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	low-latency video transmitter	low-latency video transmitter is an architecture-derived component candidate with	model:scratch_universal_architecture.drone	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
				quantity 1 candidate.				
component_candidate	generated_needs_review	generated_model	video antenna pair	video antenna pair is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.drone	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	video power filter	video power filter is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.drone	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	balance charge harness	balance charge harness is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.drone	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	charging cradle	charging cradle is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.drone	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	pack ID label	pack ID label is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.drone	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	transport storage case insert	transport storage case insert is an architecture-derived component candidate	model:scratch_universal_architecture.drone	no	yes	Review whether this is a unique physical item, shared component or separate

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
				with quantity 1 candidate.				installation instance.
component_candidate	generated_needs_review	generated_model	battery management PCB	battery management PCB is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.drone	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	high-discharge cell group	high-discharge cell group is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.drone	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.

Showing first 70 of 187 claims; export CSV for the full claim ledger.

Claim Evidence Gate

evidence_blocked

1/187 claims pass by being brief-supplied, source-backed or reviewer-accepted. 42 blocked/source-required claims, 144 review-required claims. Claim pass ratio 0.01. Blocking: 42 BoM, 185 publishable.

- architecture design has 0 blocked/source-required claim(s) and 35 review-required claim(s).
- engineering math has 2 blocked/source-required claim(s) and 3 review-required claim(s).
- bom sourcing has 40 blocked/source-required claim(s) and 102 review-required claim(s).
- compliance risk has 0 blocked/source-required claim(s) and 4 review-required claim(s).

Area	Verdict	Claims	Passed	Review Req.	Blocked	Source Req.	Blocks BoM	Blocks Publish	Pass Ratio	Required Action
brief_requirements	complete	1	1	0	0	0	0	0	1	No claim-evidence action required for this area.
architecture_design	review	35	0	35	0	0	0	35	0	Accept, correct or reject generated module and submodule allocation claims through engineering review.
engineering_math	blocked	5	0	3	2	2	2	4	0	Accept, correct or reject calculated metrics and engineering envelopes through reviewer evidence.
bom_sourcing	blocked	142	0	102	40	40	40	142	0	Admit source-backed supplier, manufacturer, MPN, unit-cost and lead-time evidence for required BoM claims.
compliance_risk	review	4	0	4	0	0	0	4	0	Accept, correct or reject class-pack regulatory and risk claims through compliance

Area	Verdict	Claims	Passed	Review Req.	Blocked	Source Req.	Blocks BoM	Blocks Publish	Pass Ratio	Required Action
										ce/domain review.

Evidence Gap Register

40 open evidence gaps: 14 blockers, 26 major, 0 minor. Closure paths: 11 sourcing intake, 21 verification intake, 7 engineering review, 0 architecture revision. Blocking rows: architecture 0, BoM 11, publishable 40.

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
readiness_blocker	blocker	open	Report readiness gate	bom score 1 is below 8.	score_repair	publishable	Admit source-backed evidence for every critical BoM line through sourcing intake. Increase priced-line coverage with admissible supplier evidence, not estimates. Collect engineering-review evidence for pending verification activities. Accept reviewer evidence for every non-sourcing verification activity before treating the report as publishable. Clear blocked verification activities or explicitly defer them with reviewer evidence. Use sourcing intake to admit source-backed supplier, manufacturer, MPN and cost evidence for critical BoM lines.
readiness_blocker	blocker	open	Report readiness gate	8 critical BoM line(s) lack source-backed cost evidence.	sourcing_intake	bom; publishable	Admit source-backed evidence for every critical BoM line through sourcing intake. Increase priced-line coverage with admissible supplier evidence, not

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
							estimates. Collect engineering-review evidence for pending verification activities. Accept reviewer evidence for every non-sourcing verification activity before treating the report as publishable. Clear blocked verification activities or explicitly defer them with reviewer evidence. Use sourcing intake to admit source-backed supplier, manufacturer, MPN and cost evidence for critical BoM lines.
readiness_blocker	blocker	open	Report readiness gate	5 verification activity/activities remain blocked.	verification_intake	publishable	Admit source-backed evidence for every critical BoM line through sourcing intake. Increase priced-line coverage with admissible supplier evidence, not estimates. Collect engineering-review evidence for pending verification activities. Accept reviewer evidence for every non-sourcing verification activity before

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
							treating the report as publishable. Clear blocked verification activities or explicitly defer them with reviewer evidence. Use sourcing intake to admit source-backed supplier, manufacturer, MPN and cost evidence for critical BoM lines.
readiness_blocker	blocker	open	Report readiness gate	0/19 non-sourcing verification activity/activities have accepted reviewer evidence.	verification_intake	publishable	Admit source-backed evidence for every critical BoM line through sourcing intake. Increase priced-line coverage with admissible supplier evidence, not estimates. Collect engineering-review evidence for pending verification activities. Accept reviewer evidence for every non-sourcing verification activity before treating the report as publishable. Clear blocked verification activities or explicitly defer them with reviewer evidence. Use sourcing intake to admit source-backed supplier,

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
							manufacturer, MPN and cost evidence for critical BoM lines.
calculation_review	major	needs review	Drone 20% reserve time	8 minutes reserved at 20% of stated endurance.	engineering_review	publishable	Confirm return-to-home reserve, hover reserve, battery ageing, wind case and payload power.
assumption_review	major	needs review	Flight endurance	Flight endurance uses duration_minutes with medium confidence.	engineering_review	publishable	Review the formula, input values and operating envelope before treating this metric as a performance claim.
assumption_review	blocker	ready for intake	Estimated CAPEX	Estimated CAPEX uses an implicit model with low confidence.	sourcing_intake	bom; publishable	Admit source-backed BoM pricing before treating this cost metric as a claim.
assumption_review	blocker	ready for intake	Estimated annual OPEX	Estimated annual OPEX uses an implicit model with low confidence.	sourcing_intake	bom; publishable	Admit source-backed BoM pricing before treating this cost metric as a claim.
assumption_review	major	needs review	Endurance Target	Endurance Target currently evaluates as warn against <=30 min typical, 30-45 min aggressive, >45 min high-risk without mass/power budget.	engineering_review	publishable	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
assumption_review	major	needs review	Quad Propulsion Allocation	Quad Propulsion Allocation currently evaluates as pass against four motors and four ESCs allocated for	engineering_review	publishable	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
				quadcopter layout.			
assumption_review	major	needs review	Control/Power /Actuation Chain	Control/Power /Actuation Chain currently evaluates as pass against power bus, motor drive, sensor bus and radio link all present.	engineering_review	publishable	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
assumption_review	major	needs review	UK-UAS	UK unmanned aircraft operating rules is relevant to this product class in UK.	engineering_review	publishable	Mass, geofencing and operating-category evidence.
assumption_review	major	needs review	CE-EMC	CE electromagnetic compatibility is relevant to this product class in UK/EU.	engineering_review	publishable	EMC test plan and declarations.
sourcing_evidence	blocker	ready for intake	Carbon airframe	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Flight battery pack	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Brushless motors	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
							time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	ESC	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Propeller set	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Flight controller	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Radio and GNSS link	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	4K camera payload	Required class-critical line must be sourced	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable,

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
				before BoM can pass.			unit cost, lead time and source quote/reference through sourcing intake.
verification_evidence	major	ready for intake	Airframe Structure And Payload Mounting	Review Airframe Structure And Payload Mounting architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Flight Energy Storage	Review Flight Energy Storage architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Flight Energy Storage	Verify Flight Energy Storage requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Propulsion And Flight Actuation	Review Propulsion And Flight Actuation architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Propulsion And Flight Actuation	Verify Propulsion And Flight Actuation requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Flight Control Compute And Communications	Review Flight Control Compute And Communications architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Flight Control Compute And Communications	Verify Flight Control Compute And Communications requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification intake template.

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
verification_evidence	major	ready for intake	Navigation And Cinematography Sensing	Review Navigation And Cinematography Sensing architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Aircraft Power Distribution	Review Aircraft Power Distribution architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Flight Safety And Containment	Review Flight Safety And Containment architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Field Maintenance And Serviceability	Review Field Maintenance And Serviceability architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Ground Operator Interface	Review Ground Operator Interface architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Flight Energy Storage -> Propulsion And Flight Actuation	Verify power_bus interface contract	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Flight Control Compute And Communications -> Propulsion And Flight Actuation	Verify motor_drive interface contract	verification_intake	publishable	Collect reviewer evidence through the verification intake template.

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
verification_evidence	major	ready for intake	Flight Control Compute And Communications -> Navigation And Cinematography Sensing	Verify sensor_bus interface contract	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	System Compliance	Plan evidence for UK-UAS	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	System Compliance	Plan evidence for CE-EMC	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	System Risk Register	Review risk control for Loss of propulsion	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	System Risk Register	Review risk control for Battery thermal event	verification_intake	publishable	Collect reviewer evidence through the verification intake template.

Closure Plan

4 active phases. Next phase: sourcing_intake. Rows by phase: architecture 0, sourcing 11, engineering 7, verification 21, score repair 1.

#	Phase	Rows	B/M/m	Blocks	Rationale	Exit Criteria	Top Gap IDs
2	Admit Source-Backed BoM Evidence	11	11/0/0	bom; publishable	Critical BoM costs, suppliers, manufacturers and MPNs must be source-backed before cost claims are credible.	Every critical BoM line has admitted supplier, manufacturer, MPN, unit cost and source reference. BoM provenance manifest has zero critical missing-source claims.	assumption:metric:capex_gbp; assumption:metric:opex_annual_gbp; readiness:2; sourcing:4k_camera_payload; sourcing:brushless_motors; sourcing:carbon_airframe; sourcing:esc; sourcing:flight_battery_pack; plus 3 more
3	Complete Engineering Review	7	0/7/0	publishable	Deterministic arithmetic can expose claims, but engineering judgment must review envelopes, assumptions and operating cases.	All calculation and assumption review rows are accepted, revised or explicitly deferred with reviewer rationale.	assumption:compliance:CE-EMC; assumption:compliance:UK-UAS; assumption:metric:headline_output; assumption:safety:drone_control_power_chain; assumption:safety:drone_endurance_target; assumption:safety:drone_propulsion_quads; calculation:drone_reserve_time_minutes
4	Accept Reviewer Evidence	21	2/19/0	publishable	Reviewer acceptance is tracked separately from deterministic scoring and must close before publication.	Every non-source verification activity is accepted, or explicitly deferred and removed from publication criteria.	readiness:3; readiness:4; verification:calculation:actuation_kinematics; verification:calculation:control_compute_communication; verification:calculation:energy_storage_source; verification:compliance_review:CE-EMC; verification:compliance_review:UK-UAS;

#	Phase	Rows	B/M/m	Blocks	Rationale	Exit Criteria	Top Gap IDs
							verification:de sign_review:a ctuation_kine matics; plus 12 more
5	Repair Remaining Section Scores	1	1/0/0	publishable	After evidence intake, rerun scoring to remove residual deterministic section blockers.	Every scored report section is at or above target and readiness verdict is publishable.	readiness:1

Depth Benchmark

prototype internal depth target. Benchmark uses aggregate counts only. Chain-v2 component names, prose, design content, prices and part numbers are not imported into the scratch generator.

3 meeting benchmark, 4 below benchmark, average comparable ratio 0.83.

Dimension	Scratch	Benchmark	Ratio	Status	Notes	Action
Functional modules	9 modules	9 modules	1	meets	Core architecture breadth.	Add missing functional modules only when the product class genuinely requires them.
Submodules	26 submodules	30 submodules	0.87	below	Submodule count is a rough proxy for engineering decomposition depth.	Expand thin modules with concrete sub-functions and carrier interfaces.
Component candidates	110 components	130 components	0.85	below	Component-word count is a rough proxy for BoM review surface, not sourcing quality.	Add justified component candidates where submodules still hide assemblies.
Interface link volume	30 links	28 links	1.07	meets	Compares link volume only; chain-v2 grammar links and scratch interface edges are not semantically identical.	Add interface contracts only where they clarify a real energy, material, signal, load or service path.
Admitted priced-line ratio	0 ratio	0.75 ratio	0	below	Prototype target expects most critical lines to become source-backed before publication.	Use sourcing intake to admit source-backed price/manufacturer/MPN evidence; do not copy benchmark estimates.
Sections scoring >=8	5 sections	6 sections	0.83	below	Readiness gate requires every scored section to reach the target.	Clear section-specific readiness actions, especially BoM evidence.
Verification activities	24 activities	20 activities	1.2	meets	Scratch-only evidence workflow depth; not compared to chain-v2 content.	Keep adding verification activities when new requirements, interfaces or standards are introduced.

Brief Requirements

Design a prosumer cinematography drone with 4K camera payload, 40 minutes flight endurance, brushless motors, ESCs, and flight controller.

Requirement	Value	Unit
Duration	40	minutes

Requirement Traceability

Requirement	Status	Value	Architecture Coverage	Evaluators	Notes
Duration	covered	40 minutes	Flight Energy Storage / Flight battery pack; Propulsion And Flight Actuation / Motor sets; Flight Control Compute And Communications / Flight controller stack	metric:headline_output; sanity:drone_endurance_target; sanity:drone_propulsion_quads; sanity:drone_control_power_chain	Endurance is traced through battery, propulsion, controls and sanity checks. Requirement has architecture coverage plus metric or engineering sanity coverage.

Requirement Coverage Gate

coverage_review_ready

1 requirement row(s): 0 accepted with evidence, 1 coverage-ready, 0 needing review, 0 blocked, 0 unlinked. Structural coverage ratio 1; accepted evidence ratio 0.

Requirement	Status	Present Signals	Missing Signals	Modules	Verification Activities	Required Action
Duration	review ready	architecture_link; submodule_link; component_link; calculation_link; review_question; verification_activity	accepted_verification	energy_storage_source; actuation_kinematics; control_compute_communication	design_review:energy_storage_source; calculation:energy_storage_source; design_review:actuation_kinematics; calculation:actuation_kinematics; design_review:control_compute_communication; calculation:control_compute_communication	Collect accepted reviewer evidence; structural requirement coverage is ready for review.

Engineering Assurance Matrix

1 requirements: 0 accepted, 0 ready, 1 needing review, 0 blocked, 0 unlinked. 1 have calculation links, 1 have review questions, 0 have accepted verification evidence.

Requirement	Overall	Architecture	Modules	Calculations	Review Questions	Verification	Blockers	Next Action
Duration	needs review	covered	Flight Energy Storage; Propulsion And Flight Actuation; Flight Control Compute And Communications	drone_endurance_hours; drone_reserve_time_minutes	assumption_resolution: component: actuation_kinematics_esc_stack_esc; assumption_resolution: component: actuation_kinematics_motor_sets_brushless_motors; assumption_resolution: component: actuation_kinematics_propeller_sets_propeller_set; assumption_resolution: component: control_compute_communication_flight_controller_stack_flight_controller; assumption_resolution: component: control_compute_communication_radio_navigation_link_radio_and_gnss_link; plus 21 more	pending; pending; pending; pending; pending	none	Complete engineering review questions, calculations or deferred verification evidence.

Compiler Stage Trace

Stage	Status	Summary	Metrics	Evidence	Limitations
Brief Parsing	passed	Extracted quantified requirements from the user brief before design generation.	requirement_count: 1; numeric_fact_count : 1; requirement_trace_covered: 1; requirement_trace_partial: 0; requirement_trace_uncovered: 0	duration_minutes: 40 minutes; duration_minutes_covered	Parsing is deterministic and currently recognises only a small set of hardware requirement patterns.
Product-Class Selection	passed	Selected drone with high confidence from brief keywords.	selected_class: drone; confidence: high; keyword_scores: drone:5	drone: 5	Classifier is keyword based; ambiguous multi-domain briefs need an LLM or richer taxonomy pass.
Universal Module Architecture	passed	Built a 9-module drone architecture from the scratch universal architecture grammar.	module_count: 9; architecture_source : scratch_universal_architecture	Airframe Structure And Payload Mounting; Flight Energy Storage; Propulsion And Flight Actuation; Flight Control Compute And Communications; Navigation And Cinematography Sensing; Aircraft Power Distribution; plus 3 more	The scratch grammar is hand-authored and deterministic; it still needs independent engineering review.
Submodule Expansion	passed	Expanded each module into engineering submodules with local purpose, interfaces and component candidates.	submodule_count: 26; average_submodules_per_module: 2.9	structure_containment.carbon_airframe ; structure_containment.payload_mounting; structure_containment.battery_mounting; energy_storage_source.flight_battery_pack; energy_storage_source.power_monitoring; energy_storage_source.charging_interface; plus 20 more	Submodule completeness is judged by deterministic coverage checks, not by a physics simulator.
Interface Graph	passed	Checked class-required module-to-module interfaces before allowing BoM review.	required_links: 3; present_links: 3; missing_links: 0; graph_nodes: 35; graph_edges: 56; shared_interface_edges: 27; missing_required_graph_edges: 0	energy_storage_source -> actuation_kinematics via power_bus: present; control_compute_communication -> actuation_kinematics via motor_drive: present;	Interfaces prove allocation coverage only; they do not yet validate ratings, thermal margins or dynamic behaviour.

Stage	Status	Summary	Metrics	Evidence	Limitations
				control_compute_communication -> sensing_instrumentation via sensor_bus: present	
Component Candidates	passed	Converted allocated component words into a candidate BoM without supplier prices.	candidate_lines: 110; critical_candidate_lines: 8; admitted_price_lines: 0	carbon_airframe: Carbon airframe; carbon_fibre_arm_tube: carbon fibre arm tube; centre_plate_pair: centre plate pair; arm_folding_hinge: arm folding hinge; landing_gear_strut: landing gear strut; camera_gimbal_mount_plate: camera gimbal mount plate; plus 24 more	Showing first 30 of 110 component candidates in the stage trace.
Architecture Readiness Gate	passed	Architecture validators found no blocker or major issue, so candidate BoM review can start.	ready_for_bom: true; blocking_issue_count: 0; sanity_pass: 2; sanity_warn: 1; sanity_fail: 0	warn Endurance Target: 40 minutes; pass Quad Propulsion Allocation: 4 motors / 4 ESCs; pass Control/Power/Actuation Chain: present	Passing this gate means deterministic architecture checks passed; it is not a certified engineering sign-off.
Sourcing And BoM Admission	warning	No supplier, manufacturer, part number or cost claims have been admitted because no source-backed evidence was provided.	admission_status: not_started; candidate_lines: 110; admitted_lines: 0; unpriced_lines: 110; unpriced_critical_lines: 8; rejected_source_records: 0	none	BoM cost remains intentionally blocked until 8 critical candidate line(s) have explicit source evidence.

Engineering Architecture Review

Ready for BoM review

9 modules, 26 sub-modules, 110 allocated component words.

From	To	Interface	Status	Engineering Reason
energy_storage_source	actuation_kinematics	power_bus	OK	Flight battery must feed propulsion loads through an explicit power bus.
control_compute_communication	actuation_kinematics	motor_drive	OK	Flight controller must command ESC/motor drive outputs.
control_compute_communication	sensing_instrumentation	sensor_bus	OK	Flight controller must receive navigation and payload state.

Interface Graph Summary

Module Nodes	Submodule Nodes	Shared Interface Edges	Required Edges	Missing Required Edges
9	26	27	3	0

Interface Contract Matrix

3/3 required contracts present. 9 shared interface families, 3 local-only declarations.

Interface	From Module	From Carrier	To Module	To Carrier	Status	Reason	Notes
power_bus	Flight Energy Storage	Flight battery pack; Power monitoring	Propulsion And Flight Actuation	Motor sets; ESC stack	present	Flight battery must feed propulsion loads through an explicit power bus.	Both endpoints expose the required interface through at least one submodule carrier.
motor_drive	Flight Control Compute And Communications	Flight controller stack	Propulsion And Flight Actuation	Motor sets; ESC stack	present	Flight controller must command ESC/motor drive outputs.	Both endpoints expose the required interface through at least one submodule carrier.
sensor_bus	Flight Control Compute And Communications	Flight controller stack; Radio and navigation link	Navigation And Cinematography Sensing	Navigation sensors; Health sensing	present	Flight controller must receive navigation and payload state.	Both endpoints expose the required interface through at least one submodule carrier.

Local-only declarations to review later: alarm_bus (Flight Safety And Containment); charger_interface (Flight Energy Storage); protective_earth (Aircraft Power Distribution).

Interface Verification Gate

interface_review_ready

3 required interface row(s): 0 accepted, 3 review-ready, 0 pending evidence, 0 deferred, 0 blocked. Structural pass ratio 1; accepted evidence ratio 0.

Interface	From	To	Carrier Status	Verification Activity	Ledger	Status	Required Action
power_bus	Flight Energy Storage	Propulsion And Flight Actuation	both_endpoints_carried	interface_review:energy_storage_source:actuation_kinematics:power_bus	pending	review_ready	Collect named interface-review evidence through verification intake.
motor_drive	Flight Control Compute And Communications	Propulsion And Flight Actuation	both_endpoints_carried	interface_review:control_compute_communication:actuation_kinematics:motor_drive	pending	review_ready	Collect named interface-review evidence through verification intake.
sensor_bus	Flight Control Compute And Communications	Navigation And Cinematography Sensing	both_endpoints_carried	interface_review:control_compute_communication:sensing_instrumentation:sensor_bus	pending	review_ready	Collect named interface-review evidence through verification intake.

Component Allocation Gate

allocation_ready

26/26 submodules ready; 0 review, 0 blocked. Component words 110; critical parts 8/8; duplicate groups 0. Allocation ratio 1; critical allocation ratio 1.

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
Airframe Structure And Payload Mounting	Carbon airframe	ready	5	motor_mounts; payload_mount; battery_mount	carbon_airframe	none	Submodule has component candidates and no allocation blocker.
	Payload mounting	ready	4	payload_mount; video_link	none	none	Submodule has component candidates and no allocation blocker.
	Battery mounting	ready	4	battery_mount; power_bus	none	none	Submodule has component candidates and no allocation blocker.
Flight Energy Storage	Flight battery pack	ready	5	power_bus; charger_interface; battery_mount	flight_battery_pack	none	Submodule has component candidates and no allocation blocker.
Flight Energy Storage	Power monitoring	ready	4	power_bus; sensor_bus	none	none	Submodule has component candidates and no allocation blocker.
Flight Energy Storage	Charging interface	ready	4	charger_interface; service_access	none	none	Submodule has component candidates and no allocation blocker.
Propulsion And Flight Actuation	Motor sets	ready	4	power_bus; motor_drive; motor_mounts	brushless_motors	none	Submodule has component candidates and no allocation blocker.
Propulsion And Flight Actuation	ESC stack	ready	5	power_bus; motor_drive; sensor_bus	esc	none	Submodule has component candidates and no allocation blocker.

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
Propulsion And Flight Actuation	Propeller sets	ready	4	motor_mounts	propeller_set	none	Submodule has component candidates and no allocation blocker.
Flight Control Compute And Communications	Flight controller stack	ready	5	sensor_bus; motor_drive; power_bus	flight_controller	none	Submodule has component candidates and no allocation blocker.
Flight Control Compute And Communications	Radio and navigation link	ready	5	radio_link; sensor_bus	radio_and_gnss_link	none	Submodule has component candidates and no allocation blocker.
Flight Control Compute And Communications	Video downlink	ready	4	video_link; power_bus	none	none	Submodule has component candidates and no allocation blocker.
Navigation And Cinematography Sensing	Camera payload	ready	5	video_link; payload_mount	4k_camera_payload	none	Submodule has component candidates and no allocation blocker.
Navigation And Cinematography Sensing	Navigation sensors	ready	4	sensor_bus	none	none	Submodule has component candidates and no allocation blocker.
Navigation And Cinematography Sensing	Health sensing	ready	4	sensor_bus; payload_mount	none	none	Submodule has component candidates and no allocation blocker.
Aircraft Power Distribution	Main power bus	ready	4	power_bus; motor_drive	none	none	Submodule has component candidates and no allocation blocker.

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
Aircraft Power Distribution	Voltage regulation	ready	4	power_bus; video_link	none	none	Submodule has component candidates and no allocation blocker.
Aircraft Power Distribution	Wiring harness	ready	4	power_bus; sensor_bus; video_link	none	none	Submodule has component candidates and no allocation blocker.
Flight Safety And Containment	Arming interlocks	ready	4	power_bus; alarm_bus	none	none	Submodule has component candidates and no allocation blocker.
Flight Safety And Containment	Lost-link response	ready	4	radio_link; sensor_bus	none	none	Submodule has component candidates and no allocation blocker.
Flight Safety And Containment	Battery safety	ready	4	power_bus; service_accesses	none	none	Submodule has component candidates and no allocation blocker.
Field Maintenance And Serviceability	Field repair access	ready	4	service_accesses; motor_mounts	none	none	Submodule has component candidates and no allocation blocker.
Field Maintenance And Serviceability	Inspection points	ready	4	service_accesses	none	none	Submodule has component candidates and no allocation blocker.
Field Maintenance And Serviceability	Transport and storage	ready	4	service_accesses; payload_mount	none	none	Submodule has component candidates and no allocation blocker.

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
Ground Operator Interface	Ground controller	ready	4	radio_link; video_link	none	none	Submodule has component candidates and no allocation blocker.
Ground Operator Interface	Mission labelling	ready	4	service_access	none	none	Submodule has component candidates and no allocation blocker.

Submodule Engineering Gate

submodule_engineering_review_required

0/26 submodules pass; 26 review, 0 blocked. Component words 110; linked requirements 3; carrier contracts 11/11; local-only interfaces 43. Review questions accepted 0/63; verification accepted 0/48; critical unpriced lines 8.

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
Airframe Structure And Payload Mounting	Carbon airframe	review	5	3	0	0/0	0/2	0/2	1	0.6667	Review 3 local-only interface declaration(s): motor_mounts, payload_mount, battery_mount. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
	Payload mounting	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): payload_mount, video_link. 2 engineering review question(s) await acceptance.

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											nce. 1 verification activity row(s) await accepted evidence.
Airframe Structure And Payload Mounting	Battery mounting	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): battery_mount, power_bus. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Flight Energy Storage	Flight battery pack	review	5	3	1	1/1	0/3	0/4	1	0.6667	Review 2 local-only interface declaration(s): charger_interface, battery_mount. 3 engineering review question(s) await acceptance. 3 verification activity

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufactur er/MPN evidence.
Flight Energy Storage	Power monitoring	review	4	2	0	1/1	0/3	0/2	0	0.6667	Review 1 local-only interface declaration(s): sensor_bus. 3 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence.
Flight Energy Storage	Charging interface	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): charger_interface, service_access. 2 engineering review questio

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											n(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Propulsion And Flight Actuation	Motor sets	review	4	3	1	2/2	0/4	0/5	1	0.6667	Review 1 local-only interface declaration(s): motor_mounts. 4 engineering review question(s) await acceptance. 4 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Propulsion And Flight Actuation	ESC stack	review	5	3	0	2/2	0/4	0/4	1	0.6667	Review 1 local-only interface declaration(s): sensor_bus. 4

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											engineering review question(s) await acceptance. 3 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Propulsion And Flight Actuation	Propellers	review	4	1	0	0/0	0/2	0/2	1	0.6667	Review 1 local-only interface declaration(s): motor_mounts. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence. 1 critical component line(s) still need

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											source-backed cost/manufacturer/MPN evidence.
Flight Control Computer And Communications	Flight controller stack	review	5	3	1	2/2	0/4	0/5	1	0.6667	Review 1 local-only interface declaration(s): power_bus. 4 engineering review question(s) await acceptance. 4 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Flight Control Computer And Communications	Radio and navigation link	review	5	2	0	1/1	0/3	0/3	1	0.6667	Review 1 local-only interface declaration(s): radio_link. 3 engineering review question(s) await acceptance.

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											nce. 2 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Flight Control Compute And Communications	Video download	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): video_link, power_bus. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Navigation And Cinematography Sensing	Camera payload	review	5	2	0	0/0	0/2	0/2	1	0.6667	Review 2 local-only interface declaration(s): video_link, payload_mount.

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Navigation And Cinematography Sensing	Navigation sensors	review	4	1	0	1/1	0/3	0/2	0	0.6667	3 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence.
Navigation And Cinematography Sensing	Health sensing	review	4	2	0	1/1	0/3	0/2	0	0.6667	Review 1 local-only interface declaration(s): payload_mount. 3 engineering review

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											question(s) await acceptance. 2 verification activity row(s) await accepted evidence.
Aircraft Power Distribution	Main power bus	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): power_bus, motor_drive. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Aircraft Power Distribution	Voltage regulation	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): power_bus, video_link. 2 engineering review question(s) await acceptance. 1 verificat

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											ion activity row(s) await accepted evidence.
Aircraft Power Distribution	Wiring harness	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): power_bus, sensor_bus, video_link. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Flight Safety And Containment	Arming interlocks	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): power_bus, alarm_bus. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											accepted evidence.
Flight Safety And Containment	Lost-link response	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): radio_link, sensor_bus. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Flight Safety And Containment	Battery safety	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): power_bus, service_access. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
Field Maintenance And Serviceability	Field repair access	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): service_access, motor_mounts. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await acceptance evidence.
Field Maintenance And Serviceability	Inspection points	review	4	1	0	0/0	0/2	0/1	0	0.6667	Review 1 local-only interface declaration(s): service_access. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await acceptance evidence.
Field Maintenance And	Transport and storage	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
Serviceability											declaration(s): service_access, payload_mount. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Ground Operator Interface	Ground controller	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): radio_link, video_link. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Ground Operator Interface	Mission labelling	review	4	1	0	0/0	0/2	0/1	0	0.6667	Review 1 local-only interface declaration(s): service_access. 2

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.

Module Engineering Gate

module_engineering_review_required

0/9 modules pass; 9 review, 0 blocked. Submodules/components: 26/110. Interface carriers 6/6; review questions accepted 0/64; critical unpriced lines 8 across 5 module(s).

Module	Verdict	Submodul es	Compone nts	Req Links	Interface s	Review Qs	Critical Unpriced	Pass Ratio	Required Action
Airframe Structure And Payload Mounting	review	3	13	0	0/0	0/5	1	0.71	1 critical BoM line(s) need source- backed cost evidence. 5 engineerin g review question(s) await acceptanc e. 1 assumptio n row(s) need review or source evidence.
Flight Energy Storage	review	3	13	1	1/1	0/11	1	0.86	1 critical BoM line(s) need source- backed cost evidence. 11 engineerin g review question(s) await acceptanc e. 6 assumptio n row(s) need review or source evidence.
Propulsion And Flight Actuation	review	3	13	1	2/2	0/14	3	0.86	3 critical BoM line(s) need source- backed cost evidence. 14 engineerin g review question(s) await acceptanc e. 9 assumptio n row(s) need review or

Module	Verdict	Submodul es	Compone nts	Req Links	Interface s	Review Qs	Critical Unpriced	Pass Ratio	Required Action
									source evidence.
Flight Control Compute And Communications	review	3	14	1	2/2	0/13	2	0.86	2 critical BoM line(s) need source-backed cost evidence. 13 engineering review question(s) await acceptance. 8 assumption row(s) need review or source evidence.
Navigation And Cinematography Sensing	review	3	13	0	1/1	0/6	1	0.71	1 critical BoM line(s) need source-backed cost evidence. 6 engineering review question(s) await acceptance. 2 assumption row(s) need review or source evidence.
Aircraft Power Distribution	review	3	12	0	0/0	0/4	0	0.71	4 engineering review question(s) await acceptance. 2 assumption row(s) need review or source evidence.
Flight Safety And	review	3	12	0	0/0	0/4	0	0.71	4 engineering review question(s)

Module	Verdict	Submodul es	Compone nts	Req Links	Interface s	Review Qs	Critical Unpriced	Pass Ratio	Required Action
Containme nt									) await acceptanc e. 1 assumptio n row(s) need review or source evidence.
Field Maintenan ce And Serviceabi lity	review	3	12	0	0/0	0/4	0	0.71	4 engineerin g review question(s)) await acceptanc e.
Ground Operator Interface	review	2	8	0	0/0	0/3	0	0.71	3 engineerin g review question(s)) await acceptanc e.

Module Review Roll-Up

4 ready, 5 sourcing-blocked, 0 needing attention.

Module	Status	Submodules	Components	Required Edges	Requirements	Critical Sourcing Blockers	Notes
Airframe Structure And Payload Mounting	sourcing	3	13	0/0	none	carbon_airframe	1 critical line(s) need source-backed evidence before BoM pass.
Flight Energy Storage	sourcing	3	13	1/1	duration_minutes	flight_battery_pack	1 critical line(s) need source-backed evidence before BoM pass.
Propulsion And Flight Actuation	sourcing	3	13	2/2	duration_minutes	brushless_motors; esc; propeller_set	3 critical line(s) need source-backed evidence before BoM pass.
Flight Control Compute And Communications	sourcing	3	14	2/2	duration_minutes	flight_controller; radio_and_gnss_link	2 critical line(s) need source-backed evidence before BoM pass.
Navigation And Cinematography Sensing	sourcing	3	13	1/1	none	4k_camera_payload	1 critical line(s) need source-backed evidence before BoM pass.
Aircraft Power Distribution	ready	3	12	0/0	none	none	No critical blockers; 12 candidate line(s) remain for later sourcing.
Flight Safety And Containment	ready	3	12	0/0	none	none	No critical blockers; 12 candidate line(s) remain for later sourcing.
Field Maintenance And Serviceability	ready	3	12	0/0	none	none	No critical blockers; 12 candidate line(s) remain for later sourcing.
Ground Operator	ready	2	8	0/0	none	none	No critical blockers; 8

Module	Status	Submodules	Components	Required Edges	Requirements	Critical Sourcing Blockers	Notes
Interface							candidate line(s) remain for later sourcing.

Engineering Review Pack

59 reviewer questions: 0 accepted, 39 ready, 20 needing review, 0 blocked. Coverage: 9 modules, 26 submodules, 3 interfaces, 2 calculations, 19 assumptions.

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
assumptio n_resoluti on	major	needs review	CE-EMC	Can the CE-EMC assumption be accepted for architecture review, or must it be replaced with evidence?	EMC test plan and declarations.	none	none	none	none
assumptio n_resoluti on	major	needs review	UK-UAS	Can the UK-UAS assumption be accepted for architecture review, or must it be replaced with evidence?	Mass, geofencing and operating-category evidence.	none	none	none	none
assumptio n_resoluti on	major	needs review	ESC	Can the ESC assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	actuation_ kinematics	none	none	none
assumptio n_resoluti on	major	needs review	Brushless motors	Can the Brushless motors assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	actuation_ kinematics	none	none	none
assumptio n_resoluti on	major	needs review	Propeller set	Can the Propeller set assumption be accepted	Admit supplier, manufacturer, MPN where applicable,	actuation_ kinematics	none	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
				for architecture review, or must it be replaced with evidence?	unit cost, lead time and evidence reference through sourcing intake.				
assumption_resolution	major	needs review	Flight controller	Can the Flight controller assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	control_compute_communication	none	none	none
assumption_resolution	major	needs review	Radio and GNSS link	Can the Radio and GNSS link assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	control_compute_communication	none	none	none
assumption_resolution	major	needs review	Flight battery pack	Can the Flight battery pack assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	energy_storage_source	none	none	none
assumption_resolution	major	needs review	4K camera payload	Can the 4K camera payload assumption be accepted for architecture review, or must it	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence	sensing_instrumentation	none	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
				be replaced with evidence?	reference through sourcing intake.				
assumption_resolution	major	needs review	Carbon airframe	Can the Carbon airframe assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	structure_containment	none	none	none
assumption_resolution	major	needs review	control_compute_communication -> actuation_kinematics	Can the control_compute_communication -> actuation_kinematics assumption be accepted for architecture review, or must it be replaced with evidence?	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.	none	motor_drive	none	none
assumption_resolution	major	needs review	control_compute_communication -> sensing_instrumentation	Can the control_compute_communication -> sensing_instrumentation assumption be accepted for architecture review, or must it be replaced with evidence?	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.	none	sensor_buses	none	none
assumption_resolution	major	needs review	energy_storage_source -> actuation_kinematics	Can the energy_storage_source -> actuation_kinematics	Review the interface carrier, direction, capacity,	none	power_buses	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
				assumption be accepted for architecture review, or must it be replaced with evidence?	safety margin and failure mode before detailed design or sourcing.				
assumption_resolution	major	needs review	Estimated CAPEX	Can the Estimated CAPEX assumption be accepted for architecture review, or must it be replaced with evidence?	Admit source-backed BoM pricing before treating this cost metric as a claim.	none	none	none	none
assumption_resolution	major	needs review	Flight endurance	Can the Flight endurance assumption be accepted for architecture review, or must it be replaced with evidence?	Review the formula, input values and operating envelope before treating this metric as a performance claim.	none	none	none	none
assumption_resolution	major	needs review	Estimated annual OPEX	Can the Estimated annual OPEX assumption be accepted for architecture review, or must it be replaced with evidence?	Admit source-backed BoM pricing before treating this cost metric as a claim.	none	none	none	none
assumption_resolution	major	needs review	Control/Power/Actuation Chain	Can the Control/Power/Actuation Chain assumption be accepted	Supply an engineering calculation, reviewer note, test plan or	energy_storage_source; actuation_kinematics; control_co	none	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				for architecture review, or must it be replaced with evidence?	datasheet-backed rationale for this envelope.	mpute_communicati on			
assumption_resoluti on	major	needs review	Endurance Target	Can the Endurance Target assumption be accepted for architecture review, or must it be replaced with evidence?	Supply an engineering calculation , reviewer note, test plan or datasheet-backed rationale for this envelope.	energy_storage_source; actuation_kinematics ; control_compute_communicati on	none	none	none
assumption_resoluti on	major	needs review	Quad Propulsion Allocation	Can the Quad Propulsion Allocation assumption be accepted for architecture review, or must it be replaced with evidence?	Supply an engineering calculation , reviewer note, test plan or datasheet-backed rationale for this envelope.	energy_storage_source; actuation_kinematics ; control_compute_communicati on	none	none	none
calculation_envelope	major	needs review	Drone 20% reserve time	Does the Drone 20% reserve time calculation support the architecture envelope, or does the concept need revision?	Confirm return-to-home reserve, hover reserve, battery ageing, wind case and payload power.	energy_storage_source; actuation_kinematics ; control_compute_communicati on	none	drone_reserve_time_minutes	none
calculation_envelope	major	ready	Drone endurance hours	Does the Drone endurance hours calculation support the architecture	Confirm payload mass, hover power, battery usable energy,	energy_storage_source; actuation_kinematics ; control_compute_co	none	drone_endurance_ho urs	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				re envelope, or does the concept need revision?	reserve margin, wind and take-off weight.	mmunicati on			
interface_contract	major	ready	Flight Control Compute And Communications -> Propulsion And Flight Actuation via motor_drive	Is the motor_drive interface between Flight Control Compute And Communications and Flight Actuation correctly owned and physically carried?	Interface-review evidence naming both endpoint owners, carrier submodules, capacity limits and any deferred checks.	control_compute_communication; actuation_kinematics	motor_drive	none	Both endpoints expose the required interface through at least one submodule carrier.
interface_contract	major	ready	Flight Control Compute And Communications -> Navigation And Cinematography Sensing via sensor_buses	Is the sensor_buses interface between Flight Control Compute And Communications and Navigation Cinematography Sensing correctly owned and physically carried?	Interface-review evidence naming both endpoint owners, carrier submodules, capacity limits and any deferred checks.	control_compute_communication; sensing_instrumentation	sensor_buses	none	Both endpoints expose the required interface through at least one submodule carrier.
interface_contract	major	ready	Flight Energy Storage -> Propulsion And Flight Actuation via power_buses	Is the power_buses interface between Flight Energy Storage and Propulsion And Flight Actuation correctly owned and physically carried?	Interface-review evidence naming both endpoint owners, carrier submodules, capacity limits and any deferred checks.	energy_storage_source; actuation_kinematics	power_buses	none	Both endpoints expose the required interface through at least one submodule carrier.

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
module_all ocation	major	ready	Propulsion And Flight Actuation	Does Propulsion And Flight Actuation have the right responsibi lity boundary, submodul e split and interface ownership for this concept?	Named engineerin g reviewer sign-off, change request or deferral note against the module allocation.	actuation_ kinematics	power_bu s; motor_driv e; motor_mo unts; sensor_bu s	drone_end urance_tar get; drone_pro pulsion_q uads; drone_con trol_power _chain	none
module_all ocation	major	ready	Flight Control Compute And Communic ations	Does Flight Control Compute And Communic ations have the right responsibi lity boundary, submodul e split and interface ownership for this concept?	Named engineerin g reviewer sign-off, change request or deferral note against the module allocation.	control_co mpute_co mmunicati on	sensor_bu s; motor_driv e; radio_link; video_link; power_bu s	drone_end urance_tar get; drone_pro pulsion_q uads; drone_con trol_power _chain	none
module_all ocation	major	ready	Flight Energy Storage	Does Flight Energy Storage have the right responsibi lity boundary, submodul e split and interface ownership for this concept?	Named engineerin g reviewer sign-off, change request or deferral note against the module allocation.	energy_st orage_sou rce	power_bu s; charger_in terface; battery_m ount; sensor_bu s; service_ac cess	drone_end urance_tar get; drone_pro pulsion_q uads; drone_con trol_power _chain	none
module_all ocation	major	ready	Ground Operator Interface	Does Ground Operator Interface have the right responsibi lity boundary, submodul e split and interface ownership	Named engineerin g reviewer sign-off, change request or deferral note against the module allocation.	hmi_ergon omics	radio_link; video_link; service_ac cess	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				for this concept?					
module_all ocation	major	ready	Field Maintenance And Serviceability	Does Field Maintenance And Serviceability have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	maintenance_service_ability	service_access; motor_mounts; payload_mount	none	none
module_all ocation	major	ready	Aircraft Power Distribution	Does Aircraft Power Distribution have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	power_distribution	power_buses; protective_earth; motor_drive; video_link; sensor_buses	none	none
module_all ocation	major	ready	Flight Safety And Containment	Does Flight Safety And Containment have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	safety_protection	radio_link; power_buses; service_access; alarm_bus; sensor_buses	none	none
module_all ocation	major	ready	Navigation And Cinematography Sensing	Does Navigation And Cinematography Sensing have the right responsibility boundary, submodule split and	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	sensing_instrumentation	sensor_buses; video_link; payload_mount	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				interface ownership for this concept?					
module_all ocation	major	ready	Airframe Structure And Payload Mounting	Does Airframe Structure And Payload Mounting have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	structure_ containme nt	motor_mo unt; payload_ mount; battery_m ount; service_ac cess; video_link; power_bu s	none	none
submodul e_allocation	major	ready	Propulsion And Flight Actuation / ESC stack	Do the componen t candidate s and interfaces assigned to ESC stack make engineerin g sense for its stated purpose?	Reviewer note confirming the submodule role, componen t allocation and any missing carriers or capacity checks.	actuation_ kinematics	power_bu s; motor_driv e; sensor_bu s	drone_end urance_tar get; drone_pro pulsion_q uads; drone_con trol_power _chain	none
submodul e_allocation	major	ready	Propulsion And Flight Actuation / Motor sets	Do the componen t candidate s and interfaces assigned to Motor sets make engineerin g sense for its stated purpose?	Reviewer note confirming the submodule role, componen t allocation and any missing carriers or capacity checks.	actuation_ kinematics	power_bu s; motor_driv e; motor_mo unt	drone_end urance_tar get; drone_pro pulsion_q uads; drone_con trol_power _chain	none
submodul e_allocation	major	ready	Propulsion And Flight Actuation / Propeller sets	Do the componen t candidate s and interfaces assigned to Propeller sets make engineerin	Reviewer note confirming the submodule role, componen t allocation and any missing	actuation_ kinematics	motor_mo unt	drone_end urance_tar get; drone_pro pulsion_q uads; drone_con trol_power _chain	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				g sense for its stated purpose?	carriers or capacity checks.				
submodul e_allocation	major	ready	Flight Control Compute And Communications / Flight controller stack	Do the componen t candidate s and interfaces assigned to Flight controller stack make engineerin g sense for its stated purpose?	Reviewer note confirming the submodul e role, componen t allocation and any missing carriers or capacity checks.	control_co mpute_co mmunicati on	sensor_bu s; motor_driv e; power_bu s	drone_end urance_tar get; drone_pro pulsion_q uads; drone_con trol_power _chain	none
submodul e_allocation	major	ready	Flight Control Compute And Communications / Radio and navigation link	Do the componen t candidate s and interfaces assigned to Radio and navigation link make engineerin g sense for its stated purpose?	Reviewer note confirming the submodul e role, componen t allocation and any missing carriers or capacity checks.	control_co mpute_co mmunicati on	radio_link; sensor_bu s	drone_end urance_tar get; drone_pro pulsion_q uads; drone_con trol_power _chain	none
submodul e_allocation	major	ready	Flight Control Compute And Communications / Video downlink	Do the componen t candidate s and interfaces assigned to Video downlink make engineerin g sense for its stated purpose?	Reviewer note confirming the submodul e role, componen t allocation and any missing carriers or capacity checks.	control_co mpute_co mmunicati on	video_link; power_bu s	drone_end urance_tar get; drone_pro pulsion_q uads; drone_con trol_power _chain	none
submodul e_allocation	major	ready	Flight Energy Storage / Charging interface	Do the componen t candidate s and interfaces assigned to Charging interface	Reviewer note confirming the submodul e role, componen t allocation and any	energy_st orage_sou rce	charger_in terface; service_ac cess	drone_end urance_tar get; drone_pro pulsion_q uads; drone_con trol_power _chain	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
submodule_allocation	major	ready	Flight Energy Storage / Flight battery pack	make engineering sense for its stated purpose?	missing carriers or capacity checks.				
				Do the component candidate s and interfaces assigned to Flight battery pack make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	energy_storage_source	power_bu s; charger_in terface; battery_m ount	drone_end urance_tar get; drone_pro pulsion_q uads; drone_con trol_power _chain	none
				Do the component candidate s and interfaces assigned to Power monitoring make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	energy_storage_source	power_bu s; sensor_bu s	drone_end urance_tar get; drone_pro pulsion_q uads; drone_con trol_power _chain	none
				Do the component candidate s and interfaces assigned to Ground controller make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	hmi_ergonomics	radio_link; video_link	none	none
				Do the component candidate s and interfaces assigned to Mission labelling make	Reviewer note confirming the submodule role, component allocation and any	hmi_ergonomics	service_ac cess	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
submodule_allocation	major	ready	Field Maintenance And Serviceability / Field repair access	engineering sense for its stated purpose?	missing carriers or capacity checks.	maintenance_serviceability	service_access; motor_mounts	none	none
				Do the component candidate s and interfaces assigned to Field repair access make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.				
				Do the component candidate s and interfaces assigned to Inspection points make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.				
				Do the component candidate s and interfaces assigned to Transport and storage make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.				
				Do the component candidate s and interfaces assigned to Aircraft Power Distribution / Main power bus make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.				
submodule_allocation	major	ready	Aircraft Power Distribution / Main power bus	Do the component candidate s and interfaces assigned	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	power_distribution	power_buses; motor_drive	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
				to Main power bus make engineering sense for its stated purpose?	t allocation and any missing carriers or capacity checks.				
submodule_allocation	major	ready	Aircraft Power Distribution / Voltage regulation	Do the component candidates and interfaces assigned to Voltage regulation make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	power_distribution	power_buses; video_link	none	none
submodule_allocation	major	ready	Aircraft Power Distribution / Wiring harness	Do the component candidates and interfaces assigned to Wiring harness make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	power_distribution	power_buses; sensor_buses; video_link	none	none
submodule_allocation	major	ready	Flight Safety And Containment / Arming interlocks	Do the component candidates and interfaces assigned to Arming interlocks make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	safety_protection	power_buses; alarm_bus	none	none
submodule_allocation	major	ready	Flight Safety And Containment / Battery safety	Do the component candidates and interfaces assigned to Battery	Reviewer note confirming the submodule role, component	safety_protection	power_buses; service_access	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
submodule_allocation	major	ready	Flight Safety And Containment / Lost-link response	safety make engineering sense for its stated purpose?	allocation and any missing carriers or capacity checks.				
				Do the component candidate s and interfaces assigned to Lost-link response make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	safety_protection	radio_link; sensor_buses	none	none
submodule_allocation	major	ready	Navigation And Cinematography Sensing / Camera payload	Do the component candidate s and interfaces assigned to Camera payload make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	sensing_instrumentation	video_link; payload_mount	none	none
submodule_allocation	major	ready	Navigation And Cinematography Sensing / Health sensing	Do the component candidate s and interfaces assigned to Health sensing make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	sensing_instrumentation	sensor_buses; payload_mount	none	none
submodule_allocation	major	ready	Navigation And Cinematography Sensing / Navigation sensors	Do the component candidate s and interfaces assigned to	Reviewer note confirming the submodule role, component	sensing_instrumentation	sensor_buses	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
				Navigation sensors make engineering sense for its stated purpose?	allocation and any missing carriers or capacity checks.				
submodule_allocation	major	ready	Airframe Structure And Payload Mounting / Battery mounting	Do the component candidate s and interfaces assigned to Battery mounting make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	structure_containment	battery_mount; power_buses	none	none
submodule_allocation	major	ready	Airframe Structure And Payload Mounting / Carbon airframe	Do the component candidate s and interfaces assigned to Carbon airframe make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	structure_containment	motor_mounts; payload_mount; battery_mount	none	none
submodule_allocation	major	ready	Airframe Structure And Payload Mounting / Payload mounting	Do the component candidate s and interfaces assigned to Payload mounting make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	structure_containment	payload_mount; video_link	none	none

Engineering Sanity Checks

Check	Status	Value	Expected	Interpretation
Endurance Target	warn	40 minutes	<=30 min typical, 30-45 min aggressive, >45 min high-risk without mass/power budget	40 minutes is an aggressive cinematography-drone target and needs a mass, propeller and battery energy budget before design freeze.
Quad Propulsion Allocation	pass	4 motors / 4 ESCs	four motors and four ESCs allocated for quadcopter layout	The architecture allocates one motor-drive channel per rotor before BoM sourcing.
Control/Power/Actuation Chain	pass	present	power bus, motor drive, sensor bus and radio link all present	Flight battery, flight controller, radio/navigation link and propulsion are connected by explicit interfaces.

Engineering Calculation Ledger

2 calculations: 1 within deterministic envelope, 1 needing engineering review, 0 outside envelope, 0 blocked by missing inputs. These are arithmetic gates, not proof of real-world performance.

Calculation	Status	Formula	Result	Envelope	Interpretation	Evidence Required
Drone endurance hours	within envelope	duration_minute s / 60	0.67 h	0.25 to 1.0 h is plausible for prosumer multirotor concepts before battery and payload proof.	0.67 h endurance converted from minutes.	Confirm payload mass, hover power, battery usable energy, reserve margin, wind and take-off weight.
Drone 20% reserve time	needs review	duration_minute s * 0.2	8 minutes	Reserve planning marker only; flight controller failsafe needs detailed energy budget.	8 minutes reserved at 20% of stated endurance.	Confirm return-to-home reserve, hover reserve, battery ageing, wind case and payload power.

Engineering Assumption Ledger

20 assumptions: 1 brief-supported, 3 model-present, 6 needing engineering review, 10 needing source evidence, 0 blocked. Architecture blockers 0; BoM blockers 10.

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
Duration	brief_requirement	brief supported	no	no	Duration = 40 minutes is treated as a design input, not as independently verified truth.	Confirm the requirement value, tolerance, duty cycle, environment and acceptance test with the project owner.
Flight endurance	derived_metric	review required	no	no	Flight endurance uses duration_minutes with medium confidence.	Review the formula, input values and operating envelope before treating this metric as a performance claim.
Estimated CAPEX	derived_metric	source required	no	yes	Estimated CAPEX uses an implicit model with low confidence.	Admit source-backed BoM pricing before treating this cost metric as a claim.
Estimated annual OPEX	derived_metric	source required	no	yes	Estimated annual OPEX uses an implicit model with low confidence.	Admit source-backed BoM pricing before treating this cost metric as a claim.
Endurance Target	sanity_envelope	review required	no	no	Endurance Target currently evaluates as warn against <=30 min typical, 30-45 min aggressive, >45 min high-risk without mass/power budget.	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
Quad Propulsion Allocation	sanity_envelope	review required	no	no	Quad Propulsion Allocation currently evaluates as pass against four motors and four ESCs allocated for quadcopter layout.	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
Control/Power/Actuation Chain	sanity_envelope	review required	no	no	Control/Power/Actuation Chain currently evaluates as pass against power bus, motor drive, sensor bus and	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
					radio link all present.	for this envelope.
energy_storage_source -> actuation_kinematics	interface_closure	model present	no	no	power_bus is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.
control_compute_communication -> actuation_kinematics	interface_closure	model present	no	no	motor_drive is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.
control_compute_communication -> sensing_instrumentation	interface_closure	model present	no	no	sensor_bus is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.
Carbon airframe	critical_component	source required	no	yes	Carbon airframe is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Flight battery pack	critical_component	source required	no	yes	Flight battery pack is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Brushless motors	critical_component	source required	no	yes	Brushless motors is a critical architecture-derived component candidate with quantity 4 each.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
ESC	critical_component	source required	no	yes	ESC is a critical architecture-derived component candidate with quantity 4 each.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Propeller set	critical_component	source required	no	yes	Propeller set is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Flight controller	critical_component	source required	no	yes	Flight controller is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Radio and GNSS link	critical_component	source required	no	yes	Radio and GNSS link is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
4K camera payload	critical_component	source required	no	yes	4K camera payload is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
UK-UAS	compliance_evidence	review required	no	no	UK unmanned aircraft operating rules is relevant to this product class in UK.	Mass, geofencing and operating-category evidence.
CE-EMC	compliance_evidence	review required	no	no	CE electromagnetic compatibility is relevant to this product class in UK/EU.	EMC test plan and declarations.

Engineering Verification Plan

24 activities: 12 ready for review, 7 open, 5 blocked.

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
Review Airframe Structure And Payload Mounting architecture allocation	ready	design_review	Airframe Structure And Payload Mounting	none	motor_mounts; payload_mount; battery_mount; service_access; video_link; power_bus	carbon_airframe; carbon_fibre_arm_tube; centre_plate_pair; arm_folding_hinge; landing_gear_strut; camera_gimbal_mount_plate; rubber_vibration_damper; payload_quick_release_latch; plus 5 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Collect source-backed evidence for Airframe Structure And Payload Mounting critical components	blocked	source_evidence	Airframe Structure And Payload Mounting	none	none	carbon_airframe	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	carbon_airframe
Review Flight Energy Storage architecture allocation	ready	design_review	Flight Energy Storage	duration_minutes	power_bus; charger_interface; battery_mount; sensor_bus;	flight_battery_pack; high_discharge_cell_group; battery_management_	Every submodule has a stated purpose and at least one component	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
					service_access	pcb; xt90_anti_s park_connector; pack_temperature_sensor; power_module_shunt; voltage_divider_board; low_voltage_alarm_output; plus 5 more	candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	
Verify Flight Energy Storage requirement calculations	open	calculation	Flight Energy Storage	duration_minutes	none	none	Each linked requirement has a calculation, sanity check or clearly stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	drone_endurance_target: 40 minutes is an aggressive cinematography-drone target and needs a mass, propeller and battery energy budget before design freeze.
Collect source-backed evidence for Flight Energy Storage critical components	blocked	source_evidence	Flight Energy Storage	duration_minutes	none	flight_battery_pack	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed quantity and	flight_battery_pack

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
							component identity match the architecture candidate line.	
Review Propulsion And Flight Actuation architecture allocation	ready	design_review	Propulsion And Flight Actuation	duration_minutes	power_bus; motor_drive; motor_mounts; sensor_bus	brushless_motors; motor_mounting_screw_set; motor_phase_lead_set; motor_bearing_shield; esc; esc_cooling_plate; pwm_signal_harness; current_telemetry_line; plus 5 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify Propulsion And Flight Actuation requirement calculations	open	calculation	Propulsion And Flight Actuation	duration_minutes	none	none	Each linked requirement has a calculation, sanity check or clearly stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	drone_endurance_target: 40 minutes is an aggressive cinematography-drone target and needs a mass, propeller and battery energy budget before design freeze.

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
Collect source-backed evidence for Propulsion And Flight Actuation critical components	blocked	source_evidence	Propulsion And Flight Actuation	duration_minutes	none	brushless_motors; esc; propeller_set	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	brushless_motors esc propeller_set
Review Flight Control Compute And Communications architecture allocation	ready	design_review	Flight Control Compute And Communications	duration_minutes	sensor_bus; motor_drive; radio_link; video_link; power_bus	flight_controller; imu_vibration_isolation_mount; barometer_foam_cover; sd_card_flight_logger; failsafe_firmware_profile; radio_and_gnss_link; gnss_antenna_mast; rc_receiver; plus 6 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify Flight Control Compute And Communications	open	calculation	Flight Control Compute And Communications	duration_minutes	none	none	Each linked requirement has a calculation, sanity check or clearly	drone_endurance_target: 40 minutes is an aggressive cinematogra

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
requirement calculations							stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	phy-drone target and needs a mass, propeller and battery energy budget before design freeze.
Collect source-backed evidence for Flight Control Compute And Communications critical components	blocked	source_evidence	Flight Control Compute And Communications	duration_minutes	none	flight_controller; radio_and_gnss_link	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	flight_controller radio_and_gnss_link
Review Navigation And Cinematography Sensing architecture allocation	ready	design_review	Navigation And Cinematography Sensing	none	sensor_bus; video_link; payload_mount	4k_camera_payload; three_axis_gimbal; camera_control_cable; lens_protection_hood; microsd_memory_card; magnetometer_module; rangefinder_altimeter; optical_flow_sensor; plus 5 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface.	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
							No blocker or major architecture issue is attached to the module.	
Collect source-backed evidence for Navigation And Cinematography Sensing critical components	blocked	source_evidence	Navigation And Cinematography Sensing	none	none	4k_camera_payload	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	4k_camera_payload
Review Aircraft Power Distribution architecture allocation	ready	design_review	Aircraft Power Distribution	none	power_bus; protective_earth; motor_drive; video_link; sensor_bus	power_distribution_pcb; copper_bus_plane; main_fuse_link; anti_spark_loop_key; 5_v_bec_regulator; 12_v_payload_regulator; lc_noise_filter; rail_status_linked; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
Review Flight Safety And Containment architecture allocation	ready	design_review	Flight Safety And Containment	none	radio_link; power_bus; service_access; alarm_bus; sensor_bus	arming_switch; status_buzzer; motor_inhibitor_logic; preflight_checklist_tag; return_to_home_failsafe; geofence_configuration; low_battery_landing_logic; flight_termination_setting; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Review Field Maintenance And Serviceability architecture allocation	ready	design_review	Field Maintenance And Serviceability	none	service_access; motor_mounts; payload_mount	spare_arm_clamp; motor_alignment_gauge; propeller_torque_tool; field_fastener_kit; crack_inspection_marker; fastener_witness_paint; battery_cycle_counter_label; payload_latch_witness_mark; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Review Ground Operator Interface architecture allocation	ready	design_review	Ground Operator Interface	none	radio_link; video_link; service_access	handheld_transmitter; ground_station_tablet; sun_hood; controller_neck_strap; aircraft_id_label; propeller_ro	Every submodule has a stated purpose and at least one component candidate. Interfaces named on	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
						tation_label; payload_mass_placard; preflight_checklist_card	the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	
Verify power_bus interface contract	ready	interface_review	Flight Energy Storage -> Propulsion And Flight Actuation	none	power_bus	flight_battery_pack; high_discharge_cell_group; battery_management_pcb; xt90_anti_spark_connector; pack_temperature_sensor; power_module_shunt; voltage_divider_board; low_voltage_alarm_output; plus 10 more	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	none
Verify motor_drive interface contract	ready	interface_review	Flight Control Compute And Communications -> Propulsion And Flight Actuation	none	motor_drive	flight_controller; imu_vibration_isolation_mount; barometer_foam_cover; sd_card_flight_logger; failsafe_firmware_profile; ; brushless_motors; motor_mounting_screw_set; motor_phase_lead_set; plus 6 more	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
Verify sensor_bus interface contract	ready	interface_review	Flight Control Compute And Communications -> Navigation And Cinematography Sensing	none	sensor_bus	flight_controller; imu_vibration_isolation_mount; barometer_foam_cover; sd_card_flight_logger; failsafe_firmware_profile; radio_and_gnss_link; gnss_antenna_mast; rc_receiver; plus 10 more	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	none
Plan evidence for UK-UAS	open	compliance_review	System Compliance	none	none	none	Mass, geofencing and operating-category evidence.	Compliance evidence not yet collected in scratch prototype.
Plan evidence for CE-EMC	open	compliance_review	System Compliance	none	none	none	EMC test plan and declarations.	Compliance evidence not yet collected in scratch prototype.
Review risk control for Loss of propulsion	open	compliance_review	System Risk Register	none	none	none	Severity, occurrence and detection ratings have been reviewed by a domain owner. Mitigation is accepted, corrected or converted into a design action. Any residual high-risk item has an owner before publication.	Risk evidence not yet reviewed in scratch prototype.
Review risk control for Battery	open	compliance_review	System Risk Register	none	none	none	Severity, occurrence and detection	Risk evidence not yet reviewed in

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
thermal event							ratings have been reviewed by a domain owner. Mitigation is accepted, corrected or converted into a design action. Any residual high-risk item has an owner before publication.	scratch prototype.

Verification Evidence Ledger

0 accepted, 0 rejected, 0 deferred, 19 pending, 0 blocked without evidence. Review coverage 0; acceptance ratio 0. Source-evidence activities remain in sourcing intake.

Activity	Ledger Status	Evidence Kind	Reviewer	Evidence Ref	Residual Action
Review Airframe Structure And Payload Mounting architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Airframe Structure And Payload Mounting critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Flight Energy Storage architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify Flight Energy Storage requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Flight Energy Storage critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Propulsion And Flight Actuation architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify Propulsion And Flight Actuation requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Propulsion And Flight Actuation critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Flight Control Compute And Communications architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify Flight Control Compute And Communications requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Flight Control Compute And Communications critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.

Activity	Ledger Status	Evidence Kind	Reviewer	Evidence Ref	Residual Action
Review Navigation And Cinematography Sensing architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Navigation And Cinematography Sensing critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Aircraft Power Distribution architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Review Flight Safety And Containment architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Review Field Maintenance And Serviceability architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Review Ground Operator Interface architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify power_bus interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.
Verify motor_drive interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.
Verify sensor_bus interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.
Plan evidence for UK-UAS	pending	compliance_review			Collect reviewer evidence through the verification intake template.
Plan evidence for CE-EMC	pending	compliance_review			Collect reviewer evidence through the verification intake template.
Review risk control for Loss of propulsion	pending	compliance_review			Collect reviewer evidence through the verification intake template.

Activity	Ledger Status	Evidence Kind	Reviewer	Evidence Ref	Residual Action
Review risk control for Battery thermal event	pending	compliance_review			Collect reviewer evidence through the verification intake template.

Airframe Structure And Payload Mounting

Carries propulsion, battery and camera payload while targeting roughly 40 minutes of flight endurance.

- motor_mounts
- payload_mount
- battery_mount
- service_access

Carbon airframe

Provides rigid arms, centre plates and landing structure.

Components: Carbon airframe, carbon fibre arm tube, centre plate pair, arm folding hinge, landing gear strut

Payload mounting

Supports vibration-isolated camera payload attachment.

Components: camera gimbal mount plate, rubber vibration damper, payload quick-release latch, payload CG adjustment rail

Battery mounting

Secures removable packs against manoeuvre loads.

Components: battery tray, battery retention strap, anti-slip battery pad, pack latch sensor

Flight Energy Storage

Stores mission energy and supplies high-current propulsion and avionics loads.

- power_bus
- charger_interface
- battery_mount
- sensor_bus

Flight battery pack

Provides swappable energy storage for flight operations.

Components: Flight battery pack, high-discharge cell group, battery management PCB, XT90 anti-spark connector, pack temperature sensor

Power monitoring

Measures pack current and voltage for endurance management.

Components: power module shunt, voltage divider board, low-voltage alarm output, battery telemetry connector

Charging interface

Supports safe pack charging and fleet rotation.

Components: balance charge harness, charging cradle, pack ID label, transport storage case insert

Propulsion And Flight Actuation

Generates lift, yaw and manoeuvre authority through motor, ESC and propeller groups.

- power_bus
- motor_drive
- motor_mounts
- sensor_bus

Motor sets

Convert electrical power into shaft torque.

Components: Brushless motors, motor mounting screw set, motor phase lead set, motor bearing shield

ESC stack

Drives motors from flight-controller commands.

Components: ESC, ESC cooling plate, PWM signal harness, current telemetry line, ESC conformal coating

Propeller sets

Generate thrust and set acoustic/efficiency envelope.

Components: Propeller set, propeller hub adapter, propeller lock nut, spare propeller pouch

Flight Control Compute And Communications

Runs stabilisation, navigation, command link and failsafe behaviours.

- sensor_bus
- motor_drive
- radio_link
- video_link
- power_bus

Flight controller stack

Stabilises aircraft and executes flight modes.

Components: Flight controller, IMU vibration isolation mount, barometer foam cover, SD card flight logger, failsafe firmware profile

Radio and navigation link

Provides pilot command, telemetry and positioning.

Components: Radio and GNSS link, GNSS antenna mast, RC receiver, telemetry radio module, antenna diversity mount

Video downlink

Streams camera status and framing view to operator.

Components: low-latency video transmitter, video antenna pair, HDMI micro cable, video power filter

Navigation And Cinematography Sensing

Captures video, inertial state and environmental data needed for controlled flight.

- sensor_bus
- video_link
- payload_mount

Camera payload

Captures stabilised 4K imagery.

Components: 4K camera payload, three-axis gimbal, camera control cable, lens protection hood, microSD media card

Navigation sensors

Measures orientation, height and heading.

Components: magnetometer module, rangefinder altimeter, optical flow sensor, airspeed estimate port

Health sensing

Monitors vibration, temperature and payload state.

Components: vibration monitor pad, payload presence switch, avionics temperature probe, landing contact switch

Aircraft Power Distribution

Routes battery power to propulsion, avionics and payload through protected low-mass paths.

- power_bus
- protective_earth
- motor_drive
- video_link

Main power bus

Distributes pack current to ESCs and avionics converters.

Components: power distribution PCB, copper bus plane, main fuse link, anti-spark loop key

Voltage regulation

Creates stable rails for flight controller, radio and payload.

Components: 5 V BEC regulator, 12 V payload regulator, LC noise filter, rail status LED

Wiring harness

Connects avionics with strain relief and field-service routing.

Components: silicone power lead set, JST signal harness, braided cable sleeve, strain relief clip set

Flight Safety And Containment

Reduces injury, runaway, battery and lost-link risks before and during flight.

- radio_link
- power_bus
- service_access
- alarm_bus

Arming interlocks

Prevents unintended motor start during handling.

Components: arming switch, status buzzer, motor inhibit logic, preflight checklist tag

Lost-link response

Defines recovery path when command link degrades.

Components: return-to-home failsafe, geofence configuration, low battery landing logic, flight termination setting

Battery safety

Manages pack handling, thermal and transport risks.

Components: fire-resistant battery pouch, pack swelling gauge, charge log sheet, thermal warning label

Field Maintenance And Serviceability

Supports rapid propeller, arm, battery and payload service between flights.

- service_access
- motor_mounts
- payload_mount

Field repair access

Makes high-wear flight parts replaceable.

Components: spare arm clamp, motor alignment gauge, propeller torque tool, field fastener kit

Inspection points

Provides visible checks before launch.

Components: crack inspection marker, fastener witness paint, battery cycle counter label, payload latch witness mark

Transport and storage

Protects aircraft between missions.

Components: foam transport case, propeller guard sleeve, gimbal transport lock, desiccant storage pouch

Ground Operator Interface

Presents mission status, camera framing and preflight instructions to the operator.

radio_link

video_link

service_access

Ground controller

Gives pilot manual control and flight status.

Components: handheld transmitter, ground-station tablet, sun hood, controller neck strap

Mission labelling

Communicates safe setup, IDs and payload status.

Components: aircraft ID label, propeller rotation label, payload mass placard, preflight checklist card

Bill of Materials

Pricing status: architecture-derived candidate BoM only. Costs, suppliers, manufacturers and part numbers have not been independently sourced.

Total CAPEX BoM: **£0**

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
Carbon airframe	1	candidate	unpriced	unpriced				
carbon fibre arm tube	1	candidate	unpriced	unpriced				
centre plate pair	1	candidate	unpriced	unpriced				
arm folding hinge	1	candidate	unpriced	unpriced				
landing gear strut	1	candidate	unpriced	unpriced				
camera gimbal mount plate	1	candidate	unpriced	unpriced				
rubber vibration damper	1	candidate	unpriced	unpriced				
payload quick-release latch	1	candidate	unpriced	unpriced				
payload CG adjustment rail	1	candidate	unpriced	unpriced				
battery tray	1	candidate	unpriced	unpriced				
battery retention strap	1	candidate	unpriced	unpriced				
anti-slip battery pad	1	candidate	unpriced	unpriced				
pack latch sensor	1	candidate	unpriced	unpriced				
Flight battery pack	1	candidate	unpriced	unpriced				
high-discharge cell group	1	candidate	unpriced	unpriced				
battery management PCB	1	candidate	unpriced	unpriced				
XT90 anti-spark connector	1	candidate	unpriced	unpriced				
pack temperature sensor	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
power module shunt	1	candidate	unpriced	unpriced				
voltage divider board	1	candidate	unpriced	unpriced				
low-voltage alarm output	1	candidate	unpriced	unpriced				
battery telemetry connector	1	candidate	unpriced	unpriced				
balance charge harness	1	candidate	unpriced	unpriced				
charging cradle	1	candidate	unpriced	unpriced				
pack ID label	1	candidate	unpriced	unpriced				
transport storage case insert	1	candidate	unpriced	unpriced				
Brushless motors	4	each	unpriced	unpriced				
motor mounting screw set	1	candidate	unpriced	unpriced				
motor phase lead set	1	candidate	unpriced	unpriced				
motor bearing shield	1	candidate	unpriced	unpriced				
ESC	4	each	unpriced	unpriced				
ESC cooling plate	1	candidate	unpriced	unpriced				
PWM signal harness	1	candidate	unpriced	unpriced				
current telemetry line	1	candidate	unpriced	unpriced				
ESC conformal coating	1	candidate	unpriced	unpriced				
Propeller set	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
propeller hub adapter	1	candidate	unpriced	unpriced				
propeller lock nut	1	candidate	unpriced	unpriced				
spare propeller pouch	1	candidate	unpriced	unpriced				
Flight controller	1	candidate	unpriced	unpriced				
IMU vibration isolation mount	1	candidate	unpriced	unpriced				
barometer foam cover	1	candidate	unpriced	unpriced				
SD card flight logger	1	candidate	unpriced	unpriced				
failsafe firmware profile	1	candidate	unpriced	unpriced				
Radio and GNSS link	1	candidate	unpriced	unpriced				
GNSS antenna mast	1	candidate	unpriced	unpriced				
RC receiver	1	candidate	unpriced	unpriced				
telemetry radio module	1	candidate	unpriced	unpriced				
antenna diversity mount	1	candidate	unpriced	unpriced				
low-latency video transmitter	1	candidate	unpriced	unpriced				
video antenna pair	1	candidate	unpriced	unpriced				
HDMI micro cable	1	candidate	unpriced	unpriced				
video power filter	1	candidate	unpriced	unpriced				
4K camera payload	1	candidate	unpriced	unpriced				
three-axis gimbal	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
camera control cable	1	candidate	unpriced	unpriced				
lens protection hood	1	candidate	unpriced	unpriced				
microSD media card	1	candidate	unpriced	unpriced				
magnetometer module	1	candidate	unpriced	unpriced				
rangefinder altimeter	1	candidate	unpriced	unpriced				
optical flow sensor	1	candidate	unpriced	unpriced				
airspeed estimate port	1	candidate	unpriced	unpriced				
vibration monitor pad	1	candidate	unpriced	unpriced				
payload presence switch	1	candidate	unpriced	unpriced				
avionics temperature probe	1	candidate	unpriced	unpriced				
landing contact switch	1	candidate	unpriced	unpriced				
power distribution PCB	1	candidate	unpriced	unpriced				
copper bus plane	1	candidate	unpriced	unpriced				
main fuse link	1	candidate	unpriced	unpriced				
anti-spark loop key	1	candidate	unpriced	unpriced				
5 V BEC regulator	1	candidate	unpriced	unpriced				
12 V payload regulator	1	candidate	unpriced	unpriced				
LC noise filter	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
rail status LED	1	candidate	unpriced	unpriced				
silicone power lead set	1	candidate	unpriced	unpriced				
JST signal harness	1	candidate	unpriced	unpriced				
braided cable sleeve	1	candidate	unpriced	unpriced				
strain relief clip set	1	candidate	unpriced	unpriced				
arming switch	1	candidate	unpriced	unpriced				
status buzzer	1	candidate	unpriced	unpriced				
motor inhibit logic	1	candidate	unpriced	unpriced				
preflight checklist tag	1	candidate	unpriced	unpriced				
return-to-home failsafe	1	candidate	unpriced	unpriced				
geofence configuration	1	candidate	unpriced	unpriced				
low battery landing logic	1	candidate	unpriced	unpriced				
flight termination setting	1	candidate	unpriced	unpriced				
fire-resistant battery pouch	1	candidate	unpriced	unpriced				
pack swelling gauge	1	candidate	unpriced	unpriced				
charge log sheet	1	candidate	unpriced	unpriced				
thermal warning label	1	candidate	unpriced	unpriced				
spare arm clamp	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
motor alignment gauge	1	candidate	unpriced	unpriced				
propeller torque tool	1	candidate	unpriced	unpriced				
field fastener kit	1	candidate	unpriced	unpriced				
crack inspection marker	1	candidate	unpriced	unpriced				
fastener witness paint	1	candidate	unpriced	unpriced				
battery cycle counter label	1	candidate	unpriced	unpriced				
payload latch witness mark	1	candidate	unpriced	unpriced				
foam transport case	1	candidate	unpriced	unpriced				
propeller guard sleeve	1	candidate	unpriced	unpriced				
gimbal transport lock	1	candidate	unpriced	unpriced				
desiccant storage pouch	1	candidate	unpriced	unpriced				
handheld transmitter	1	candidate	unpriced	unpriced				
ground-station tablet	1	candidate	unpriced	unpriced				
sun hood	1	candidate	unpriced	unpriced				
controller neck strap	1	candidate	unpriced	unpriced				
aircraft ID label	1	candidate	unpriced	unpriced				
propeller rotation label	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
payload mass placard	1	candidate	unpriced	unpriced				
preflight checklist card	1	candidate	unpriced	unpriced				

Cost And Sourcing

Sourcing Admission

Status	Candidate Lines	Admitted Prices	Unpriced Lines	Unpriced Critical Lines
not_started	110	0	110	8

Critical Sourcing Worklist

Component	Qty	Reason
Carbon airframe	1 candidate	Required class-critical line must be sourced before BoM can pass.
Flight battery pack	1 candidate	Required class-critical line must be sourced before BoM can pass.
Brushless motors	4 each	Required class-critical line must be sourced before BoM can pass.
ESC	4 each	Required class-critical line must be sourced before BoM can pass.
Propeller set	1 candidate	Required class-critical line must be sourced before BoM can pass.
Flight controller	1 candidate	Required class-critical line must be sourced before BoM can pass.
Radio and GNSS link	1 candidate	Required class-critical line must be sourced before BoM can pass.
4K camera payload	1 candidate	Required class-critical line must be sourced before BoM can pass.

No rejected source records.

BoM Sourcing Ledger

0/110 priced lines admitted. Critical coverage 0/8. Rejected evidence records: 0. Cost/manufacturer/MPN fields stay blank until evidence is admitted.

Component identity check: no duplicate componentWorldId groups detected.

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
Carbon airframe	critical unpriced	unique	critical	unpriced					Collect admissible supplier/m anufacture r/MPN/unit -cost evidence before BoM can pass.
carbon fibre arm tube	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
centre plate pair	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
arm folding hinge	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
landing gear strut	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
camera gimbal mount plate	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									planning starts.
rubber vibration damper	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
payload quick-release latch	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
payload CG adjustment rail	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
battery tray	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
battery retention strap	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
anti-slip battery pad	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									procurement planning starts.
pack latch sensor	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
Flight battery pack	critical unpriced	unique	critical	unpriced					Collect admissible supplier/m anufacturer/MPN/unit -cost evidence before BoM can pass.
high-discharge cell group	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
battery management PCB	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
XT90 anti-spark connector	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
pack temperature sensor	candidate unpriced	unique	candidate	unpriced					Source after critical lines or

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									when detailed procurement planning starts.
power module shunt	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
voltage divider board	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
low-voltage alarm output	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
battery telemetry connector	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
balance charge harness	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
charging cradle	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
pack ID label	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
transport storage case insert	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
Brushless motors	critical unpriced	unique	critical	unpriced					Collect admissible supplier/m anufacturer/MPN/unit -cost evidence before BoM can pass.
motor mounting screw set	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
motor phase lead set	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									planning starts.
motor bearing shield	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
ESC	critical unpriced	unique	critical	unpriced					Collect admissible supplier/m anufacturer/MPN/unit -cost evidence before BoM can pass.
ESC cooling plate	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
PWM signal harness	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
current telemetry line	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
ESC conformal coating	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									procurement planning starts.
Propeller set	critical unpriced	unique	critical	unpriced					Collect admissible supplier/m anufacture r/MPN/unit -cost evidence before BoM can pass.
propeller hub adapter	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
propeller lock nut	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
spare propeller pouch	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
Flight controller	critical unpriced	unique	critical	unpriced					Collect admissible supplier/m anufacture r/MPN/unit -cost evidence before BoM can pass.

Showing first 40 of 110 ledger rows; export CSV for the full line-by-line ledger.

0 source-backed claims, 32 missing-source claims, 0 provenance violations. Critical missing-source claims: 32. Supplier, manufacturer, MPN and cost fields must appear here with source refs before procurement use.

Line	Claim Field	Status	Value	Source Ref	Next Action
Carbon airframe	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Carbon airframe	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Carbon airframe	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Carbon airframe	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Flight battery pack	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Flight battery pack	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Flight battery pack	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Flight battery pack	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Brushless motors	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Brushless motors	manufacturer	missing source			Admit source-backed

Line	Claim Field	Status	Value	Source Ref	Next Action
					supplier/manufactur er/MPN/unit-cost evidence before BoM can pass.
Brushless motors	mpn	missing source			Admit source- backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass.
Brushless motors	unit_cost_gbp	missing source			Admit source- backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass.
ESC	supplier	missing source			Admit source- backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass.
ESC	manufacturer	missing source			Admit source- backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass.
ESC	mpn	missing source			Admit source- backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass.
ESC	unit_cost_gbp	missing source			Admit source- backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass.
Propeller set	supplier	missing source			Admit source- backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass.
Propeller set	manufacturer	missing source			Admit source- backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass.
Propeller set	mpn	missing source			Admit source- backed supplier/manufactur er/MPN/unit-cost

Line	Claim Field	Status	Value	Source Ref	Next Action
					evidence before BoM can pass.
Propeller set	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Flight controller	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Flight controller	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Flight controller	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Flight controller	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Radio and GNSS link	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Radio and GNSS link	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Radio and GNSS link	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Radio and GNSS link	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.

Line	Claim Field	Status	Value	Source Ref	Next Action
4K camera payload	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
4K camera payload	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
4K camera payload	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
4K camera payload	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.

Source Reference Quality Gate

No sourcing evidence references have been admitted. Manufacturer, MPN and cost fields must remain blank until source-backed evidence is admitted through sourcing intake.

BoM Evidence Trace Matrix

0/110 BoM row(s) are production-eligible; 0 protocol-only, 0 source-reference blocked, 8 critical unsourced. Priced-review rows 0; procurement use no; next row bom-trace:structure_containment_carbon_airframe_carbon_airframe.

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
Carbon airframe	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
carbon fibre arm tube	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
centre plate pair	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
arm folding hinge	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
landing gear strut	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
camera gimbal mount plate	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									or procurement planning starts.
rubber vibration damper	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
payload quick-release latch	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
payload CG adjustment rail	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
battery tray	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
battery retention strap	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									planning starts.
anti-slip battery pad	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
pack latch sensor	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
Flight battery pack	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
high-discharge cell group	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
battery management PCB	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
XT90 anti-spark connector	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									critical sourcing is complete or procurement planning starts.
pack temperature sensor	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
power module shunt	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
voltage divider board	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
low-voltage alarm output	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
battery telemetry connector	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									procurement planning starts.
balance charge harness	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
charging cradle	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
pack ID label	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
transport storage case insert	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
Brushless motors	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
motor mounting screw set	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
motor phase lead set	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
motor bearing shield	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
ESC	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
ESC cooling plate	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
PWM signal harness	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									or procurement planning starts.
current telemetry line	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
ESC conformal coating	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
Propeller set	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
propeller hub adapter	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
propeller lock nut	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
spare propeller pouch	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
Flight controller	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
IMU vibration isolation mount	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
barometer foam cover	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
SD card flight logger	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
failsafe firmware profile	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									or procurement planning starts.
Radio and GNSS link	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
GNSS antenna mast	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
RC receiver	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
telemetry radio module	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
antenna diversity mount	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
low-latency video transmitter	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

Showing first 50 of 110 BoM trace rows; export CSV for the full trace.

BoM Evidence Closure Plan

110 closure row(s): 8 ready, 0 blocked, 102 deferred. Actions: 8 collect source, 0 repair, 0 replace protocol, 102 candidate deferrals. Procurement blockers 8; priced-review blockers 8; next row bom-close:actuation_kinematics_esc_stack_esc.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
1	ESC	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
2	Brushless motors	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
3	Propeller set	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
4	Flight controller	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
5	Radio and GNSS link	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
6	Flight battery pack	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
7	4K camera payload	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
8	Carbon airframe	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
9	current telemetry line	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
10	ESC conformal coating	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
11	ESC cooling plate	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
12	PWM signal harness	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is	no	no	Keep as candidate-only until critical sourcing is complete or

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
						promoted to procurement scope			procurement planning starts.
13	motor bearing shield	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
14	motor mounting screw set	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
15	motor phase lead set	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
16	propeller hub adapter	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
17	propeller lock nut	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to	no	no	Keep as candidate-only until critical sourcing is complete or procurement

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
						procurement scope			planning starts.
18	spare propeller pouch	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
19	barometer foam cover	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
20	failsafe firmware profile	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
21	IMU vibration isolation mount	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
22	SD card flight logger	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
23	antenna diversity mount	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
24	GNSS antenna mast	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
25	RC receiver	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
26	telemetry radio module	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
27	HDMI micro cable	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
28	low-latency video transmitter	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
29	video antenna pair	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
30	video power filter	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
31	balance charge harness	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
32	charging cradle	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
33	pack ID label	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
34	transport storage case insert	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
35	battery management PCB	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
36	high-discharge cell group	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
37	pack temperature sensor	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
38	XT90 anti-spark connector	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
39	battery telemetry connector	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
40	low-voltage alarm output	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
41	power module shunt	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
42	voltage divider board	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
43	controller neck strap	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
44	ground-station tablet	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
45	handheld transmitter	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
46	sun hood	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
47	aircraft ID label	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
48	payload mass placard	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
49	preflight checklist card	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
50	propeller rotation label	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

Showing first 50 of 110 BoM closure rows; export CSV for the full queue.

Sourcing Batch Plan

2 batch(es): 1 active, 0 waiting, 1 deferred. Active rows 8; deferred rows 102. Critical source rows 8; repair rows 0; protocol replacements 0. Next batch sourcing-batch:2:critical_source_collection; next item critical_source_collection:1:esc.

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
2	Collect Critical Source Evidence	active	blocker	ESC	ESC; ESC manufacturer part number; ESC catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	Brushless motors	Brushless motors; Brushless motors manufacturer part number; Brushless motors catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	Propeller set	Propeller set; Propeller set manufacturer part number; Propeller set catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	Flight controller	Flight controller; Flight controller manufacturer part number; Flight controller catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	Radio and GNSS link	Radio and GNSS link; Radio and GNSS link manufacturer part number; Radio and GNSS link catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
2	Collect Critical Source Evidence	active	blocker	Flight battery pack	Flight battery pack; Flight battery pack manufacturer part number; Flight battery pack catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	4K camera payload	4K camera payload; 4K camera payload manufacturer part number; 4K camera payload catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	Carbon airframe	Carbon airframe; Carbon airframe manufacturer part number; Carbon airframe catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
6	Defer Candidate-Only Sourcing	deferred	minor	current telemetry line	current telemetry line; current telemetry line manufacturer part number; current telemetry line catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	ESC conformal coating	ESC conformal coating; ESC conformal coating manufacturer part number; ESC conformal	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
					coating catalogue price GBP			
6	Defer Candidate-Only Sourcing	deferred	minor	ESC cooling plate	ESC cooling plate; ESC cooling plate manufacturer part number; ESC cooling plate catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	PWM signal harness	PWM signal harness; PWM signal harness manufacturer part number; PWM signal harness catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	motor bearing shield	motor bearing shield; motor bearing shield manufacturer part number; motor bearing shield catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	motor mounting screw set	motor mounting screw set; motor mounting screw set manufacturer part number; motor mounting screw set catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	motor phase lead set	motor phase lead set; motor phase lead set	no immediate evidence required unless this	no	candidate remains out of procurement totals

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
					manufacturer part number; motor phase lead set catalogue price GBP	candidate is promoted to procurement scope		
6	Defer Candidate-Only Sourcing	deferred	minor	propeller hub adapter	propeller hub adapter; propeller hub adapter manufacturer part number; propeller hub adapter catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	propeller lock nut	propeller lock nut; propeller lock nut manufacturer part number; propeller lock nut catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	spare propeller pouch	spare propeller pouch; spare propeller pouch manufacturer part number; spare propeller pouch catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	barometer foam cover	barometer foam cover; barometer foam cover manufacturer part number; barometer foam cover catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-	deferred	minor	failsafe firmware	failsafe firmware	no immediate	no	candidate remains out

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
	Only Sourcing			profile	profile; failsafe firmware profile manufacturer part number; failsafe firmware profile catalogue price GBP	evidence required unless this candidate is promoted to procurement scope		of procurement totals

Procurement Readiness Gate

procurement_not_started

0/7 procurement areas pass; 2 review, 5 blocked. Critical production-eligible BoM rows 0/8; procurement-blocking closure rows 8; active sourcing rows 8; deferred sourcing rows 102. Source quality no_sourcing_evidence; evidence authenticity no_evidence; costing costing_not_started; BoM admission candidate_bom_review_required. Can use for procurement: no. Next action: Collect production-quality source evidence for critical BoM rows before procurement use..

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture review	review	0.17	architecture_freeze_review_required; structurally ready yes, independent review accepted no.	none	Collect accepted engineering/interface/requirement review evidence before procurement use.
bom evidence trace	blocked	0	0/8 critical rows production-eligible; 8 critical unsourced, 0 protocol-only, 0 source-reference blocked.	structure_containment_carbon_airframe_carbon_airframe: Collect admissible source evidence for this critical line before BoM can pass. energy_storage_source_flight_battery_pack_flight_battery_pack: Collect admissible source evidence for this critical line before BoM can pass. actuation_kinematics_motor_sets_brushless_motors: Collect admissible source evidence for this critical line before BoM can pass. actuation_kinematics_esc_stack_esc: Collect admissible source evidence for this critical line before BoM can pass. actuation_kinematics_propeller_sets_propeller_set: Collect admissible source evidence for this critical line before BoM can pass. plus 3 more	Collect production-quality source evidence for critical BoM rows before procurement use.
sourcing batch closure	blocked	0.9273	8 procurement-blocking closure row(s); 8 active sourcing row(s), 102 deferred row(s).	bom-close:actuation_kinematics_esc_stack_esc: Collect admissible source evidence for this critical line before BoM can pass. bom-close:actuation_kinematics_motor_sets_brushless_motors: Collect admissible source evidence for this critical line before BoM can pass. bom-close:actuation_kinematics_esc_stack_esc: Collect admissible source evidence for this critical line before BoM can pass.	Work the active sourcing batches until procurement-blocking closure rows are closed.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
				ematics_propeller_sets_propeller_set: Collect admissible source evidence for this critical line before BoM can pass. bom-close:control_compute_communication_flight_controller_stack_flight_controller: Collect admissible source evidence for this critical line before BoM can pass. bom-close:control_compute_communication_radio_navigation_link_radio_and_gnss_link: Collect admissible source evidence for this critical line before BoM can pass. plus 3 more	
source reference quality	blocked	0	no_sourcing_evidence; 0/0 source-reference row(s) pass; 0 protocol, 0 placeholder.	none	Admit production source evidence before procurement readiness can start.
evidence authenticity	blocked	0	no_evidence; 0/0 evidence rows production-ready; 0 protocol.	No evidence records are present.	Add source and reviewer evidence before procurement readiness can start.
bom costing	blocked	0.14	costing_not_started ; critical priced 0/8; BoM total 0 GBP.	structure_containment_carbon_airframe_carbon_airframe: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. energy_storage_source_flight_battery_pack_flight_battery_pack: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. actuation_kinematics_motor_sets_brushless_motors: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. actuation_kinematics_esc_stack_esc: Collect admissible	Close costing blockers before procurement use.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
				supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. actuation_kinematics_propeller_sets_propeller_set: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. plus 46 more	
prototype procurement policy	review	0	candidate_bom_review_required; display mode candidate_only; prototype procurement use allowed no.	none	Prototype policy keeps procurement use disabled until external procurement review and approval are explicitly added.

Sourcing Evidence Pack

Evidence packets request source data only. They do not admit supplier, manufacturer, MPN or cost claims.

Critical Line	Qty	Search Starts	Required Fields	Acceptance Criteria
Carbon airframe	1 candidate	Carbon airframe; Carbon airframe manufacturer part number; Carbon airframe catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
Flight battery pack	1 candidate	Flight battery pack; Flight battery pack manufacturer part number; Flight battery pack catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
Brushless motors	4 each	Brushless motors; Brushless motors manufacturer part number; Brushless motors catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
ESC	4 each	ESC; ESC manufacturer part number; ESC catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.

Critical Line	Qty	Search Starts	Required Fields	Acceptance Criteria
Propeller set	1 candidate	Propeller set; Propeller set manufacturer part number; Propeller set catalogue price GBP	componentWorldId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWorldId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
Flight controller	1 candidate	Flight controller; Flight controller manufacturer part number; Flight controller catalogue price GBP	componentWorldId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWorldId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
Radio and GNSS link	1 candidate	Radio and GNSS link; Radio and GNSS link manufacturer part number; Radio and GNSS link catalogue price GBP	componentWorldId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWorldId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
4K camera payload	1 candidate	4K camera payload; 4K camera payload manufacturer part number; 4K camera payload catalogue price GBP	componentWorldId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWorldId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.

Cost model status: pending sourcing. Architecture is being reviewed before cost claims are admitted.

Cost Bucket	Estimate
CAPEX	£0
Annual OPEX	£0
NRE	£25,000

No admitted supplier spend yet. Supplier rows appear only after source-backed evidence records are admitted.

Regulatory And Risk

Standard	Title	Jurisdiction	Evidence Required
UK-UAS	UK unmanned aircraft operating rules	UK	Mass, geofencing and operating-category evidence.
CE-EMC	CE electromagnetic compatibility	UK/EU	EMC test plan and declarations.

Hazard	S	O	D	RPN	Mitigation
Loss of propulsion	9	3	5	135	Motor health monitoring and conservative payload limits.
Battery thermal event	8	3	4	96	Pack protection, charging limits and enclosure venting.

Section Issues And Score

bom

- blocker / critical_part_unpriced: Carbon airframe is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Flight battery pack is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Brushless motors is critical but has no credible unit cost.
- blocker / critical_part_unpriced: ESC is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Propeller set is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Flight controller is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Radio and GNSS link is critical but has no credible unit cost.
- blocker / critical_part_unpriced: 4K camera payload is critical but has no credible unit cost.
- major / low_priced_line_ratio: Only 0% of BoM lines have prices.

Section Score Ledger

Rule: start at 9, subtract 3 per blocker, 1.5 per major, 0.5 per minor, floor at 1. Mean score 7.67. This is a deterministic gate score from current validators, not an external reviewer score or proof of real-world engineering quality. Reviewer acceptance is tracked separately in the verification ledger.

Section	Base	B/M/m Issues	Deduction	Final	Target	Issue Deductions	Rationale
executive_summary	9	0/0/0	-0	9	pass	none	No current deterministic validator issue is attached to this scored section.
brief_requirements	9	0/0/0	-0	9	pass	none	No current deterministic validator issue is attached to this scored section.
design_modules	9	0/0/0	-0	9	pass	none	No current deterministic validator issue is attached to this scored section.
bom	9	8/1/0	-25.5	1	below target	blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); major/low_priced_line_ratio (-1.5)	Score is base minus deterministic deductions for attached section issues.
feasibility_notes	9	0/0/0	-0	9	pass	none	No current deterministic validator issue is attached to this scored section.
sources_references	9	0/0/0	-0	9	pass	none	No current deterministic validator issue is attached to this scored section.

Section	Base	B/M/m Issues	Deduction	Final	Target	Issue Deductions	Rationale
Mean	7.67						